

Optimizing IT Support Team Performance Using Analytics (SUPPORTLYTICS)

Presented by :

Dharani Sree M

MILESTONE - 1

MODULE 1 & MODULE 2

Project Background:

Organizations receive thousands of IT support requests every day through different communication channels. These support tickets contain valuable information regarding:

- Customer issues
- Product-related problems
- Ticket priorities
- Resolution performance
- Customer satisfaction

However, without proper analysis, organizations face challenges such as:

- Increased ticket resolution time
- Repeated customer complaints
- Poor workload distribution
- SLA violations
- Reduced customer satisfaction

Objectives

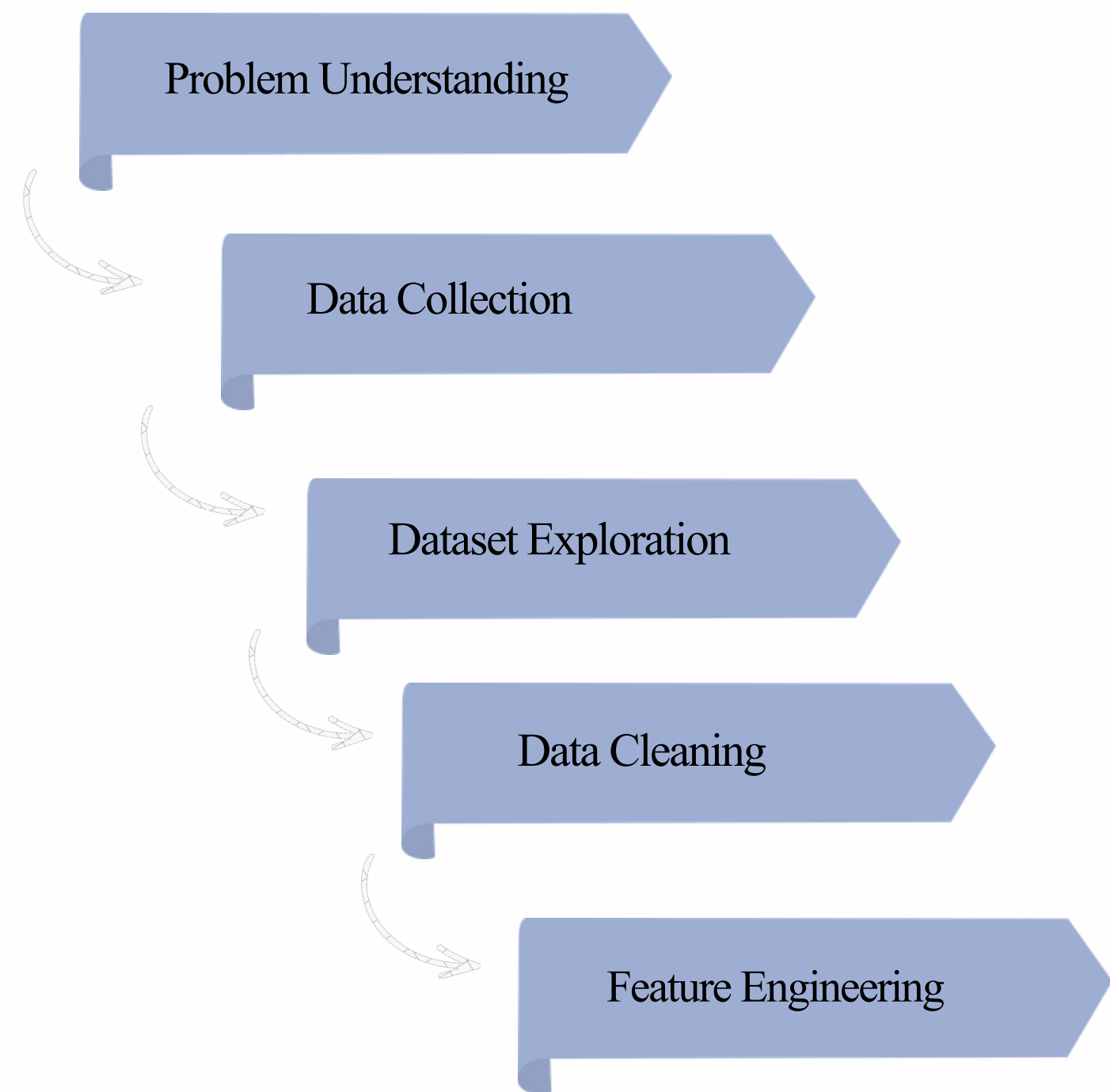
The objective of the Supportlytics project is to analyze IT support ticket data and convert raw operational data into meaningful business insights using analytics and visualization techniques. The project aims to improve support efficiency, monitor key performance indicators, and assist organizations in making data-driven decisions.

Scope of the Project

The project includes:

- Data Collection
- Data Cleaning
- Exploratory Data Analysis
- KPI Analysis
- Dashboard Development
- Business Recommendations

WORKFLOW:



KPI’S:

KPI	PURPOSE	DATASET COLUMN
Average Resolution Time	Measures how quickly tickets are resolved	resolution_time_hours
Total Tickets	Measures total support requests received	ticket_id
Tickets by Status	Shows Open, Closed, and In Progress tickets	status
Tickets by priority	Identifies workload based on urgency	priority
Tickets by issue Type	Identifies the most common technical issues	issue_type
Tickets by product area	Shows which products generate more tickets	product_area
Tickets by region	Identifies geographic distribution of tickets	region
Tickets by platform	Shows which platforms experience more issues	platform
Reopened tickets	Measures recurring or unresolved problems	reopened
CSAT	Evaluates customer satisfaction	csat_score
Customer Sentiment	Analyzes customer feedback trends	customer_sentiment

DATA UNDERSTANDING:

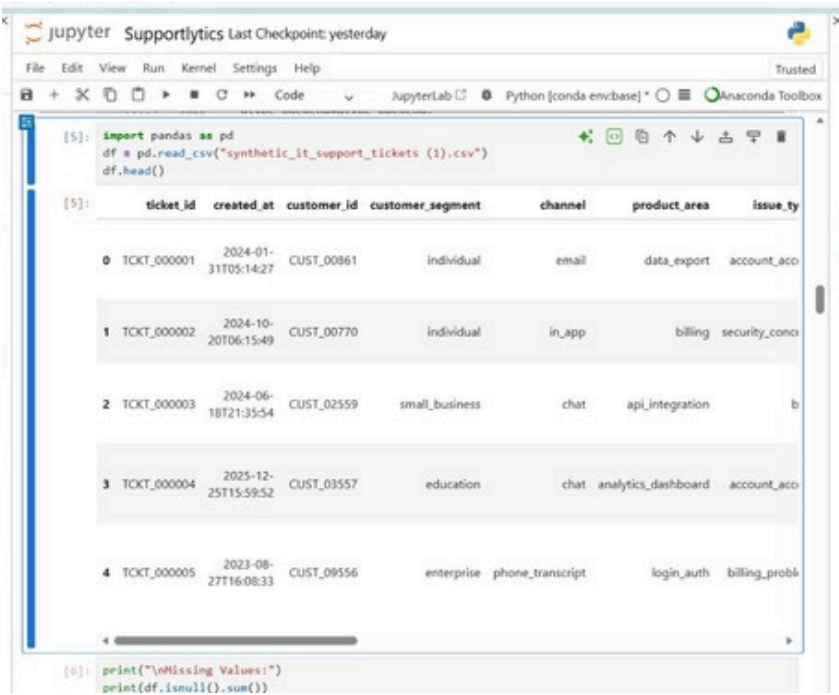
After loading the dataset into Jupyter Notebook, I explored its structure to understand the available information and identify data quality issues.

DATASET INFORMATION:

Attribute	value
Total Records	100,000
Total Columns	20
Object Columns	16
Numeric Columns	4

Activities Performed

- Loaded the dataset using Pandas.
- Examined dataset dimensions.
- Viewed column names.
- Verified data types.
- Generated descriptive statistics.
- Counted unique values.
- Identified missing values.
- Verified duplicate records.



DATASET ANALYSIS:

In [10]:

df.describe(include="all")

Out[10]:

	ticket_id	created_at	customer_id	customer_segment	channel	product_area	issue_type	priority	status	sla_plan	initial_message	agent_first_re
count	100000	100000	100000	100000	100000	100000	100000	100000	100000	100000	100000	1000
unique	100000	99959	9999	5	5	7	8	4	5	3	96	
top	TCKT_000001	2025-10-23T07:27:06	CUST_07153	small_business	phone_transcript	mobile_app	how_to	low	resolved	standard	Something seems off, can you please check my w...	Happy to h with yc question. We v loo
freq	1	2	26	20203	20133	14432	12739	39735	50131	60066	4279	50
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	N

In [7]:

print(df.columns)

Index(['ticket_id', 'created_at', 'customer_id', 'customer_segment', 'channel', 'product_area', 'issue_type', 'priority', 'status', 'sla_plan', 'initial_message', 'agent_first_reply', 'resolution_summary', 'resolution_time_hours', 'reopened', 'customer_sentiment', 'csat_score', 'has_attachment', 'platform', 'region'], dtype='object')

In [8]:

df.describe()

Out[8]:

	resolution_time_hours	reopened	csat_score	has_attachment
count	60113.000000	100000.000000	100000.000000	100000.000000
mean	45.013123	0.050450	2.241950	0.249520
std	49.202797	0.218873	1.771599	0.432737
min	0.110000	0.000000	0.000000	0.000000
25%	14.880000	0.000000	0.000000	0.000000
50%	29.910000	0.000000	2.000000	0.000000
75%	51.660000	0.000000	4.000000	0.000000
max	239.960000	1.000000	5.000000	1.000000

In [11]:

df.dtypes

Out[11]:

ticket_id	object
created_at	object
customer_id	object
customer_segment	object
channel	object
product_area	object
issue_type	object
priority	object
status	object
sla_plan	object
initial_message	object
agent_first_reply	object
resolution_summary	object
resolution_time_hours	float64
reopened	int64
customer_sentiment	object
csat_score	int64
has_attachment	int64
platform	object
region	object
dtype:	object

In [12]:

1 created_at 100000 non-null object

2 customer_id 100000 non-null object

3 customer_segment 100000 non-null object

4 channel 100000 non-null object

5 product_area 100000 non-null object

6 issue_type 100000 non-null object

7 priority 100000 non-null object

8 status 100000 non-null object

9 sla_plan 100000 non-null object

10 initial_message 100000 non-null object

11 agent_first_reply 100000 non-null object

12 resolution_summary 60113 non-null object

13 resolution_time_hours 60113 non-null float64

14 reopened 100000 non-null int64

15 customer_sentiment 100000 non-null object

16 csat_score 100000 non-null int64

17 has_attachment 100000 non-null int64

18 platform 100000 non-null object

19 region 80003 non-null object

dtypes: float64(1), int64(3), object(16)

memory usage: 15.3+ MB

In [6]:

print("Rows and Columns:", df.shape)

Rows and Columns: (100000, 20)

In [7]:

print(df.columns)

Index(['ticket_id', 'created_at', 'customer_id', 'customer_segment', 'channel', 'product_area', 'issue_type', 'priority', 'status', 'sla_plan', 'initial_message', 'agent_first_reply', 'resolution_summary', 'resolution_time_hours', 'reopened', 'customer_sentiment', 'csat_score', 'has_attachment', 'platform', 'region'], dtype='object')

DATA PREPROCESSING:

Missing Value Analysis

➤ Identified missing values in Region, Resolution Summary, and Resolution Time Hours

Missing Value Handling

➤ Filled missing values using appropriate methods (Unknown, Not Resolved Yet, 0)

Duplicate Verification

➤ Checked duplicate records

]Data Validation

➤ Verified data consistency after cleaning
Prepare data for analytics

Result

- Improved dataset quality.
- Removed incomplete information.
- Prepared data for feature engineering.

```
In [14]: df.isnull().sum()

Out[14]: ticket_id      0
         created_at     0
         customer_id    0
         customer_segment 0
         channel        0
         product_area    0
         issue_type      0
         priority        0
         status          0
         sla_plan        0
         initial_message 0
         agent_first_reply 0
         resolution_summary 39887
         resolution_time_hours 39887
         reopened        0
         customer_sentiment 0
         csat_score      0
         has_attachment   0
         platform        0
         region          19997
         dtype: int64

In [15]: df["resolution_summary"] = df["resolution_summary"].fillna("Not Resolved Yet")

In [16]: df["resolution_time_hours"] = df["resolution_time_hours"].fillna(0)

In [17]: df["region"] = df["region"].fillna("Unknown")
```

FEATURE ENGINEERIN & INITIAL ANALYSIS:

Feature Created:

Resolution_Duration

- Represents the time required to resolve each support ticket.
- Supports resolution time analysis and SLA monitoring.

```
In [33]: df["Resolution_Duration"] = df["resolution_time_hours"]
```

```
In [34]: df[["resolution_time_hours", "Resolution_Duration"]].head()
```

	resolution_time_hours	Resolution_Duration
0	36.53	36.53
1	238.32	238.32
2	0.00	0.00
3	0.00	0.00
4	61.32	61.32

Priority_Score

Converted ticket priorities into numerical values:

- Low → 1
- Medium → 2
- High → 3
- Urgent → 4

```
In [35]: priority_mapping = {
          "low":1,
          "medium":2,
          "high":3,
          "urgent":4
        }

df["Priority_Score"] = df["priority"].map(priority_mapping)
```

```
In [36]: df[["priority", "Priority_Score"]].head(10)
```

	priority	Priority_Score
0	low	1
1	medium	2
2	low	1
3	medium	2
4	low	1
5	low	1
6	urgent	4
7	low	1
8	high	3
9	medium	2

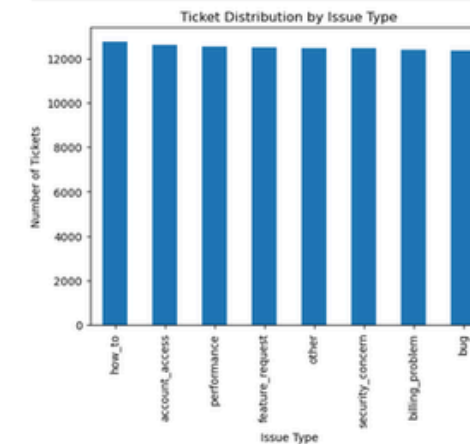
INITIAL TICKET DISTRIBUTION ANALYSIS PERFORMED:

- Analyzed ticket distribution by Priority to understand the workload across Low, Medium, High, and Urgent tickets.
- Examined ticket distribution by Product Area (Category) to identify the products or services generating the highest number of support requests.
- Studied ticket distribution by Issue Type to identify the most frequent customer issues and recurring support problems.

Key Findings:

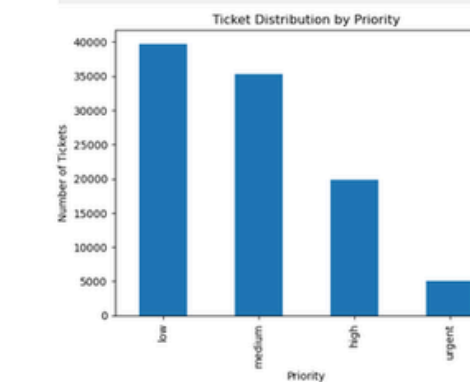
- ✓ Understood the overall distribution of support tickets.
- ✓ Identified major ticket categories and priority trends.
- ✓ Prepared the dataset for detailed Exploratory Data Analysis (EDA) and KPI evaluation.

```
plt.title("Ticket Distribution by Issue Type")
plt.xlabel("Issue Type")
plt.ylabel("Number of Tickets")
plt.show()
```



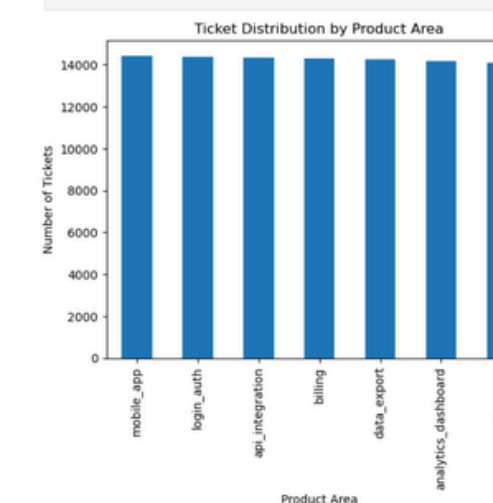
Ticket Distribution by issue type

```
In [25]: import matplotlib.pyplot as plt
df["priority"].value_counts().plot(kind="bar")
plt.title("Ticket Distribution by Priority")
plt.xlabel("Priority")
plt.ylabel("Number of Tickets")
plt.show()
```



Ticket Distribution by priority

```
plt.title("Ticket Distribution by Product Area")
plt.xlabel("Product Area")
plt.ylabel("Number of Tickets")
plt.show()
```



Ticket Distribution by Category

CONCLUSION & OUTCOMES:

Key Outcomes:

- Successfully loaded and explored the IT support ticket dataset.
- Understood the dataset structure and validated data quality.
- Identified and handled missing values.
- Verified that no duplicate records were present.
- Created new analytical features (Resolution_Duration and Priority_Score).
- Saved the cleaned dataset for further analysis.
- Uploaded the notebook and cleaned dataset to the assigned GitHub branch.

MILESTONE - 2

MODULE 3 & MODULE 4

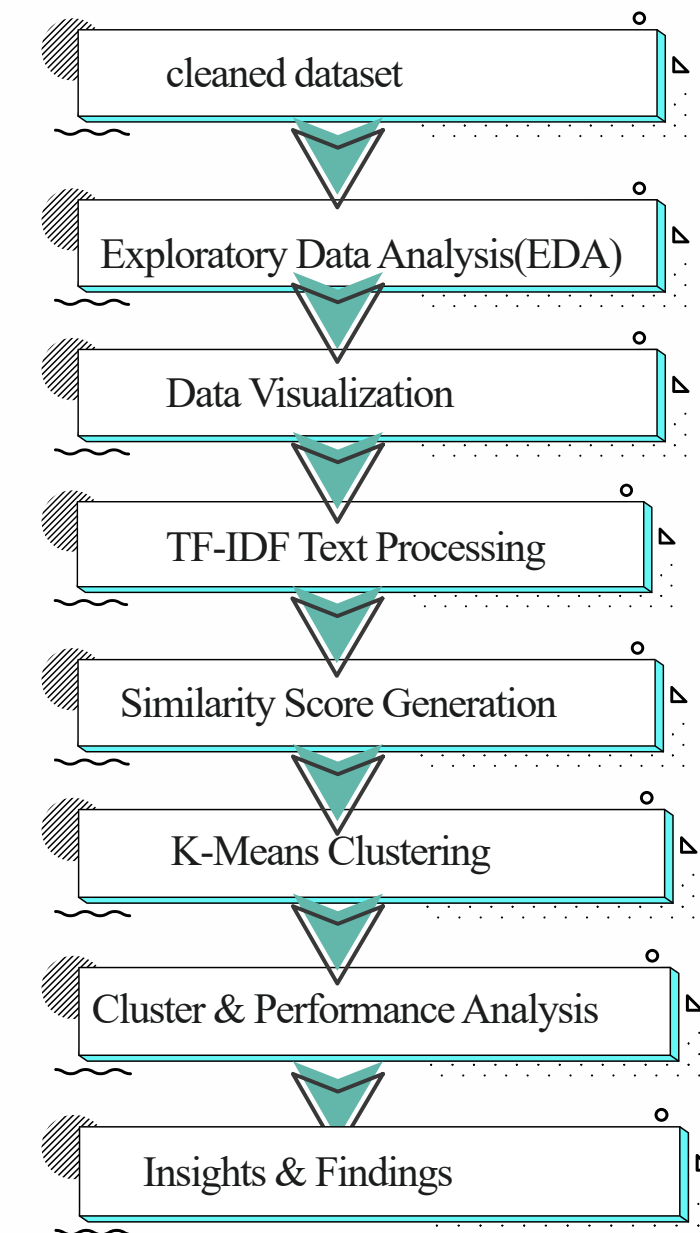
OBJECTIVES:

In this milestone, I focused on exploring the cleaned dataset in greater detail and applying machine learning techniques to identify ticket patterns.

ACTIVITIES PERFORMED:

- Performed Exploratory Data Analysis (EDA)
- Visualized ticket distributions using different charts
- Generated Similarity Scores using TF-IDF
- Applied K-Means Clustering
- Analyzed cluster insights
- Compared cluster performance using statistical visualizations

WORKFLOW:



MODULE 3: EXPLORATORY VISUALIZATION

OBJECTIVE:

To understand the characteristics and distribution of IT support tickets using exploratory visual analysis.

Activities Performed:

- ✓ Ticket Type Distribution
- ✓ Category Distribution
- ✓ Priority Distribution
- ✓ Queue Distribution

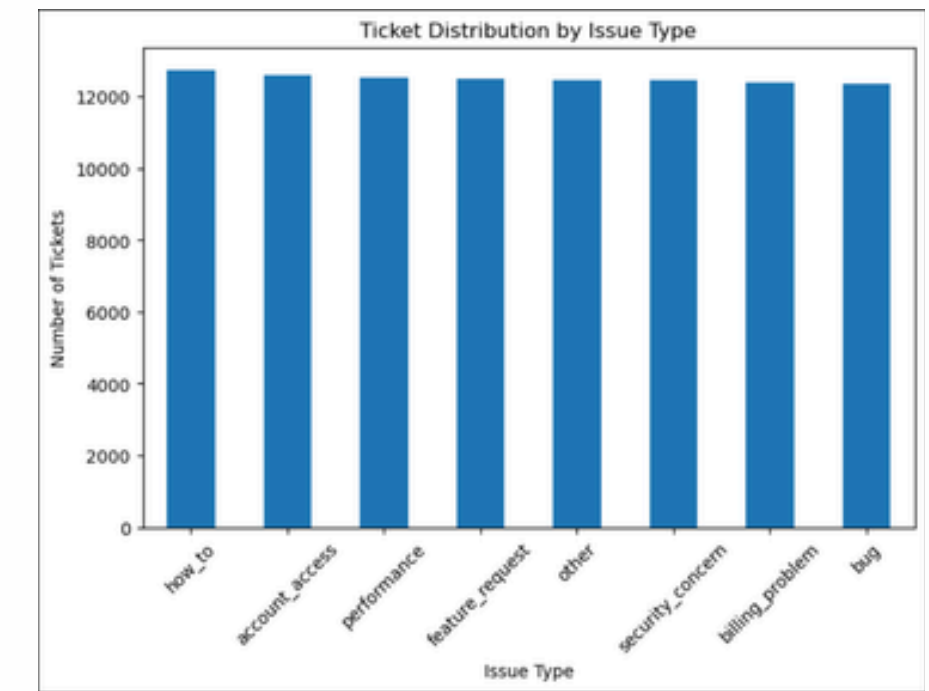
TICKET TYPE DISTRIBUTION:

OBJECTIVE:

To identify the most frequently reported support issues.

METHODOLOGY:

- ➔ Analyzed the Issue Type column.
- ➔ Calculated the frequency of each issue.
- ➔ Created a bar chart using Matplotlib



```
df["issue_type"].value_counts()

issue_type
how_to      12739
account_access  12609
performance  12513
feature_request  12487
other        12463
security_concern  12458
billing_problem  12382
bug          12349
Name: count, dtype: int64

df["issue_type"].value_counts().plot(kind="bar", figsize=(8,5))

plt.title("Ticket Distribution by Issue Type")
plt.xlabel("Issue Type")
plt.ylabel("Number of Tickets")
plt.xticks(rotation=45)

plt.show()
```

CATEGORY DISTRIBUTION:

OBJECTIVE:

To identify the categories receiving the highest number of support requests.

METHODOLOGY:

Renamed Product Area to Category.

Counted category frequencies.

Visualized category distribution.

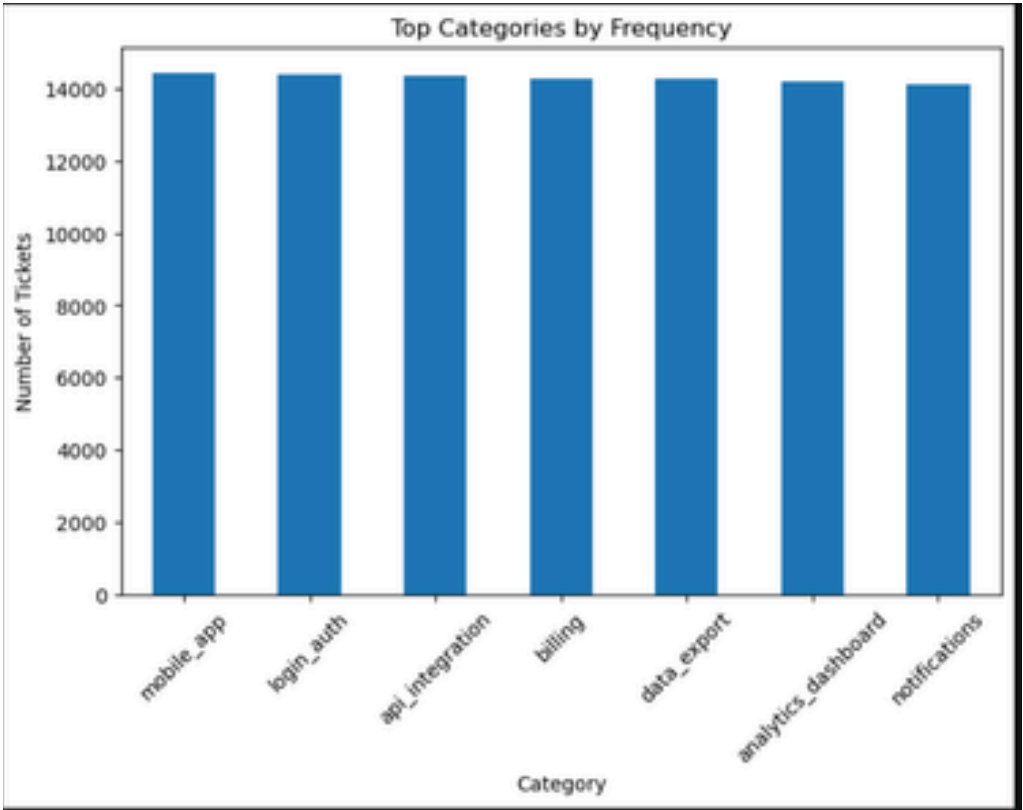
```
df["category"].value_counts()

category
mobile_app      14432
login_auth      14381
api_integration  14350
billing          14289
data_export     14262
analytics_dashboard 14187
notifications    14099
Name: count, dtype: int64

df["category"].value_counts().plot(kind="bar", figsize=(8,5))

plt.title("Top Categories by Frequency")
plt.xlabel("Category")
plt.ylabel("Number of Tickets")
plt.xticks(rotation=45)

plt.show()
```



PRIORITY & QUEUE ANALYSIS:

OBJECTIVE:

To understand ticket allocation based on priority and support teams.

METHODOLOGY:

- Calculated ticket count by priority.
- Calculated ticket count by queue.
- Compared workload across queues.

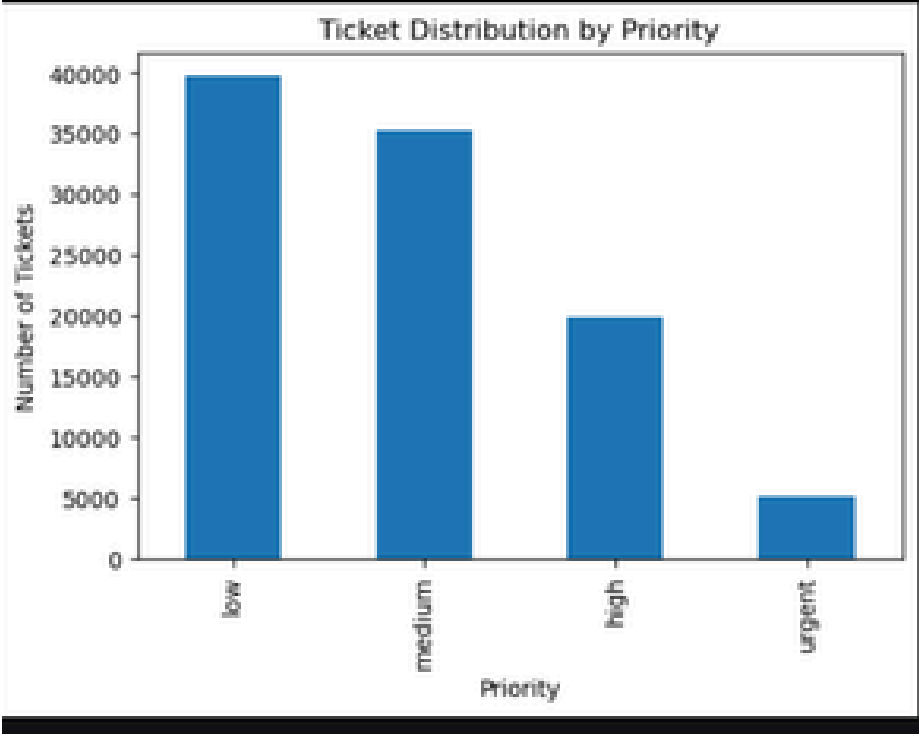
```
df["priority"].value_counts()
```

```
[11]
--
priority
low      39735
medium   35313
high     19865
urgent    5087
Name: count, dtype: int64
```

```
df["priority"].value_counts().plot(kind="bar", figsize=(6,4))

plt.title("Ticket Distribution by Priority")
plt.xlabel("Priority")
plt.ylabel("Number of Tickets")

plt.show()
```



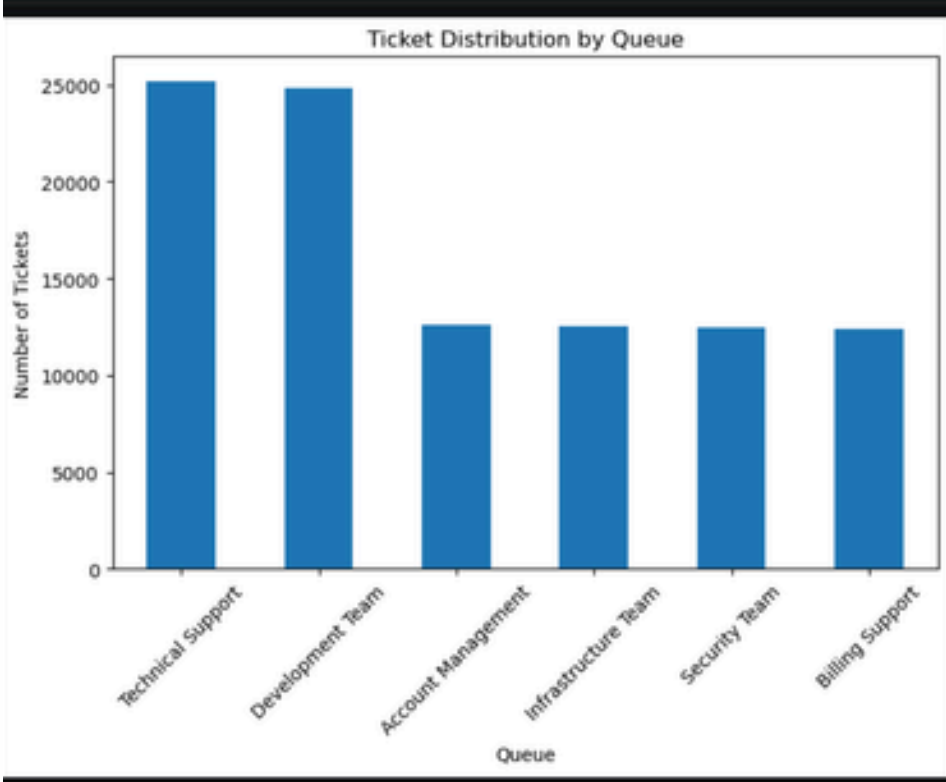
```
queue_map = {
    "account_access": "Account Management",
    "billing_problem": "Billing Support",
    "security_concern": "Security Team",
    "bug": "Development Team",
    "performance": "Infrastructure Team",
    "feature_request": "Development Team",
    "how_to": "Technical Support",
    "other": "Technical Support"
}

df["Queue"] = df["issue_type"].map(queue_map)

df["Queue"].value_counts().plot(kind="bar", figsize=(8,5))

plt.title("Ticket Distribution by Queue")
plt.xlabel("Queue")
plt.ylabel("Number of Tickets")
plt.xticks(rotation=45)

plt.show()
```



MODULE 4: SIMILARITY & CLUSTER INSIGHTS

OBJECTIVE:

To group similar support tickets using text processing and machine learning techniques.

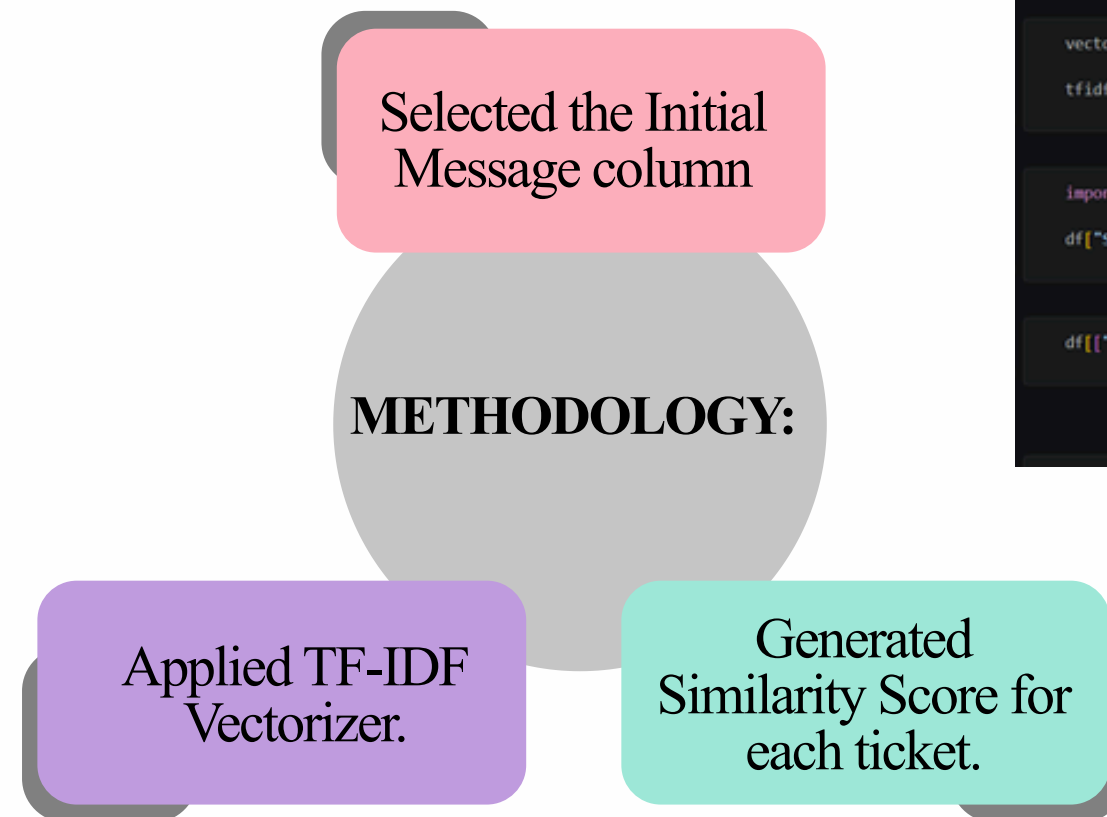
Activities Performed:

- ✓ TF-IDF Feature Extraction
- ✓ Similarity Score Generation
- ✓ K-Means Clustering
- ✓ Cluster Analysis
- ✓ Performance Visualization

TF-IDF & SIMILARITY SCORE:

OBJECTIVE:

To convert textual ticket descriptions into numerical features suitable for machine learning.



```
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(df["initial_message"])

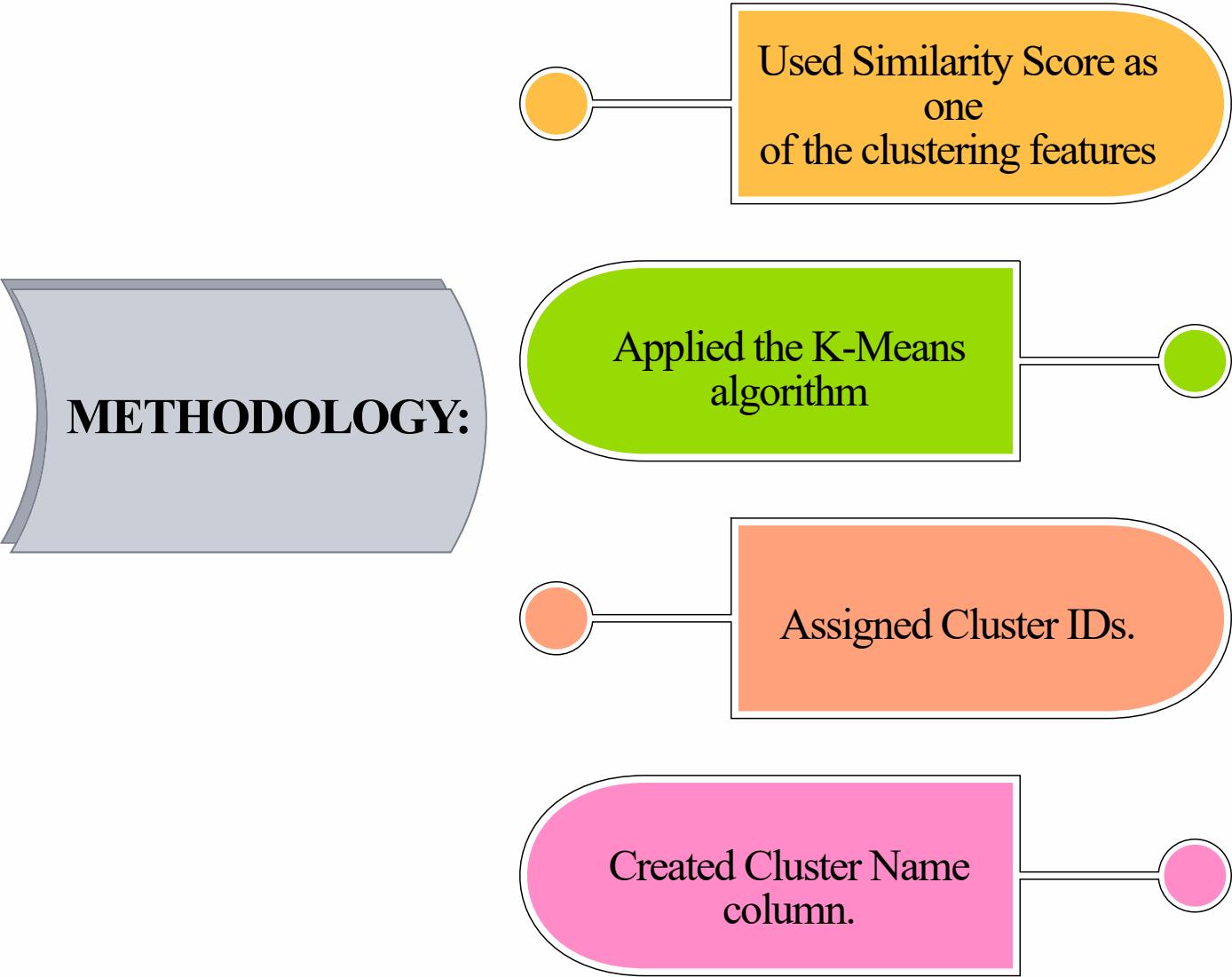
import numpy as np
df["Similarity_Score"] = np.asarray(tfidf_matrix.mean(axis=1)).ravel()

df[["Similarity_Score"]].head()
```

K - MEANS CLUSTERING:

OBJECTIVE:

To group tickets with similar characteristics into meaningful clusters.



```
[19] from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

df["category_encoded"] = encoder.fit_transform(df["category"])
df["issue_type_encoded"] = encoder.fit_transform(df["issue_type"])
df["priority_encoded"] = encoder.fit_transform(df["priority"])

[20] features = df[
    [
        "category_encoded",
        "issue_type_encoded",
        "priority_encoded",
        "Similarity_Score"
    ]
]

[21] from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_features = scaler.fit_transform(features)
```

```
[22] from sklearn.cluster import KMeans

kmeans = KMeans(
    n_clusters=4,
    random_state=42,
    n_init=10
)

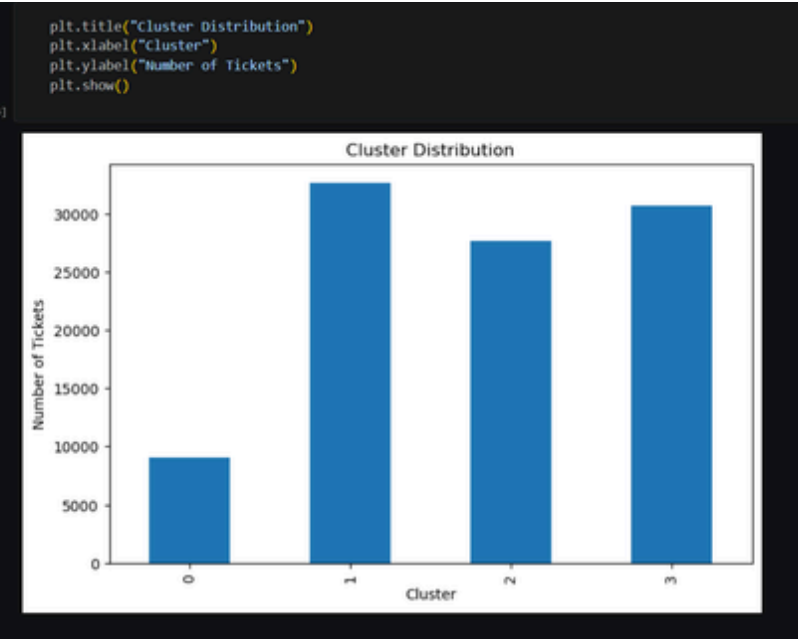
df["Cluster"] = kmeans.fit_predict(scaled_features)

[23] df["Cluster"].value_counts()

Cluster
1    32615
3    30672
2    27644
0     9069
Name: count, dtype: int64

[24] df["Cluster"].value_counts().sort_index().plot(kind="bar", figsize=(8,5))

plt.title("Cluster Distribution")
plt.xlabel("Cluster")
plt.ylabel("Number of Tickets")
plt.show()
```



CLUSTER INSIGHTS:

OBJECTIVE:

To analyze cluster characteristics and dominant issue types.

METHODOLOGY:

Compared cluster size.

Compared issue type frequency within each cluster.

Calculated average similarity score.

```
df.groupby("Cluster")["Similarity_Score"].mean()

Cluster
0    0.017115
1    0.021599
2    0.021690
3    0.021684
Name: Similarity_Score, dtype: float64

pd.crosstab(df["Cluster"], df["issue_type"])

pd.crosstab(df["Cluster"], df["issue_type"]).plot(
    kind="bar",
    figsize=(10,5)
)

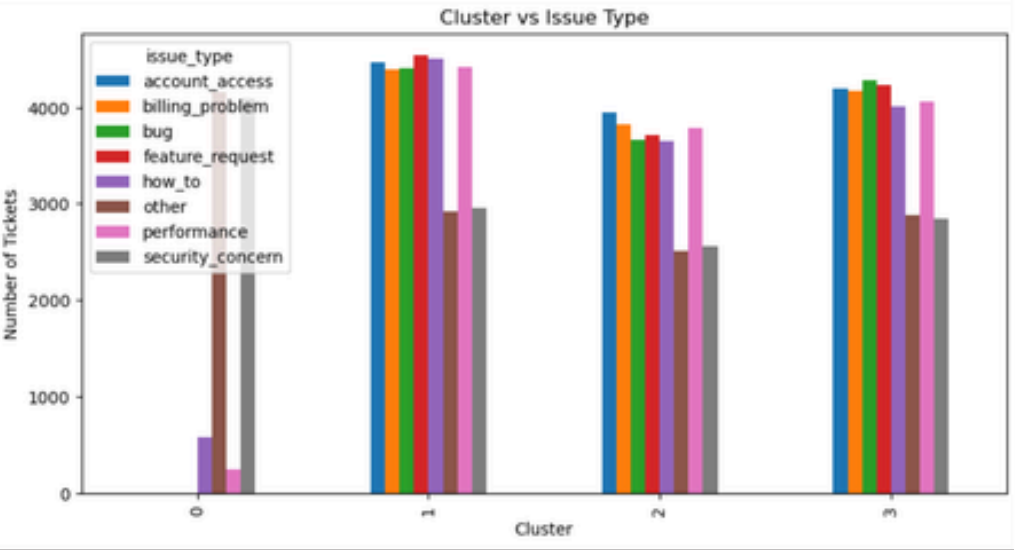
plt.title("Cluster vs Issue Type")
plt.xlabel("Cluster")
plt.ylabel("Number of Tickets")
plt.show()
```

```
plt.figure(figsize=(8,6))

df.boxplot(
    column="resolution_time_hours",
    by="Cluster",
    showfliers=False
)

plt.title("Resolution Time by Cluster")
plt.suptitle("")
plt.xlabel("Cluster")
plt.ylabel("Resolution Time (Hours)")

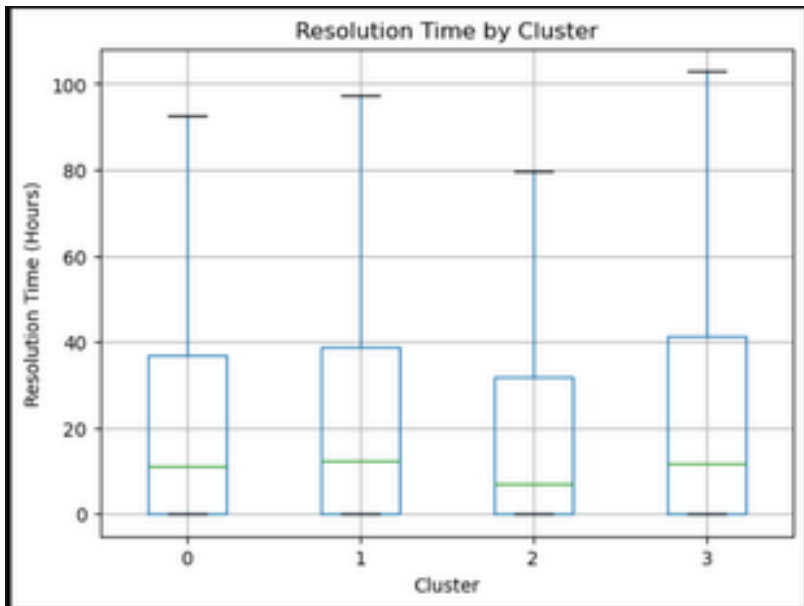
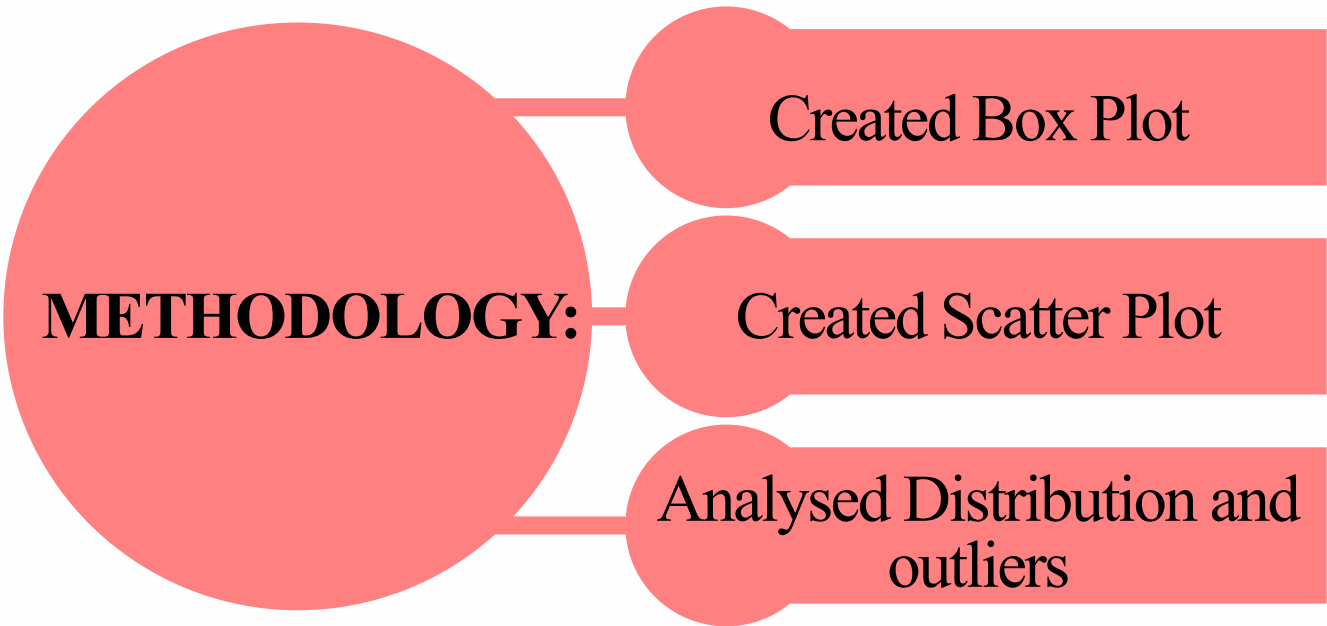
plt.show()
```



PERFORMANCE VISUALIZATION:

OBJECTIVE:

To evaluate variations in ticket performance.



```
plt.figure(figsize=(8,6))

df.boxplot(
    column="resolution_time_hours",
    by="Cluster",
    showfliers=False
)

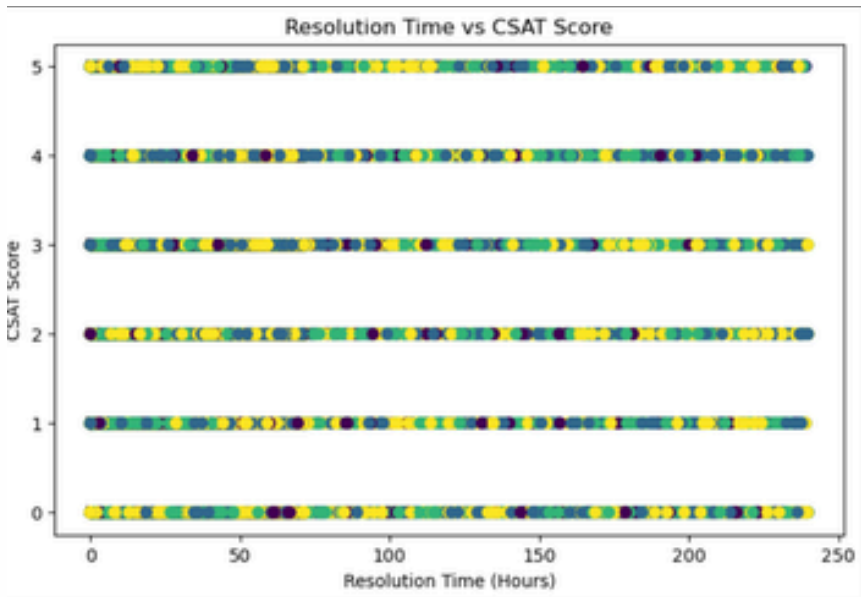
plt.title("Resolution Time by Cluster")
plt.suptitle("")
plt.xlabel("Cluster")
plt.ylabel("Resolution Time (Hours)")

plt.show()
```

```
plt.figure(figsize=(8,5))

plt.scatter(
    df["resolution_time_hours"],
    df["csat_score"],
    c=df["Cluster"]
)

plt.xlabel("Resolution Time (Hours)")
plt.ylabel("CSAT Score")
plt.title("Resolution Time vs CSAT Score")
plt.show()
```



KEY ACHIEVEMENTS:

Milestone Deliverables:

- ✓ Exploratory Data Analysis Completed
- ✓ Eight Visualizations Created
- ✓ Similarity Score Generated
- ✓ TF-IDF Implemented
- ✓ K-Means Clustering Completed
- ✓ Cluster Insights Generated
- ✓ Performance Analysis Completed

CONCLUSION:

Milestone 2 successfully transformed the cleaned support ticket dataset into meaningful analytical insights through exploratory visualization and machine learning.

MILESTONE - 3

MODULE 5 & MODULE 6

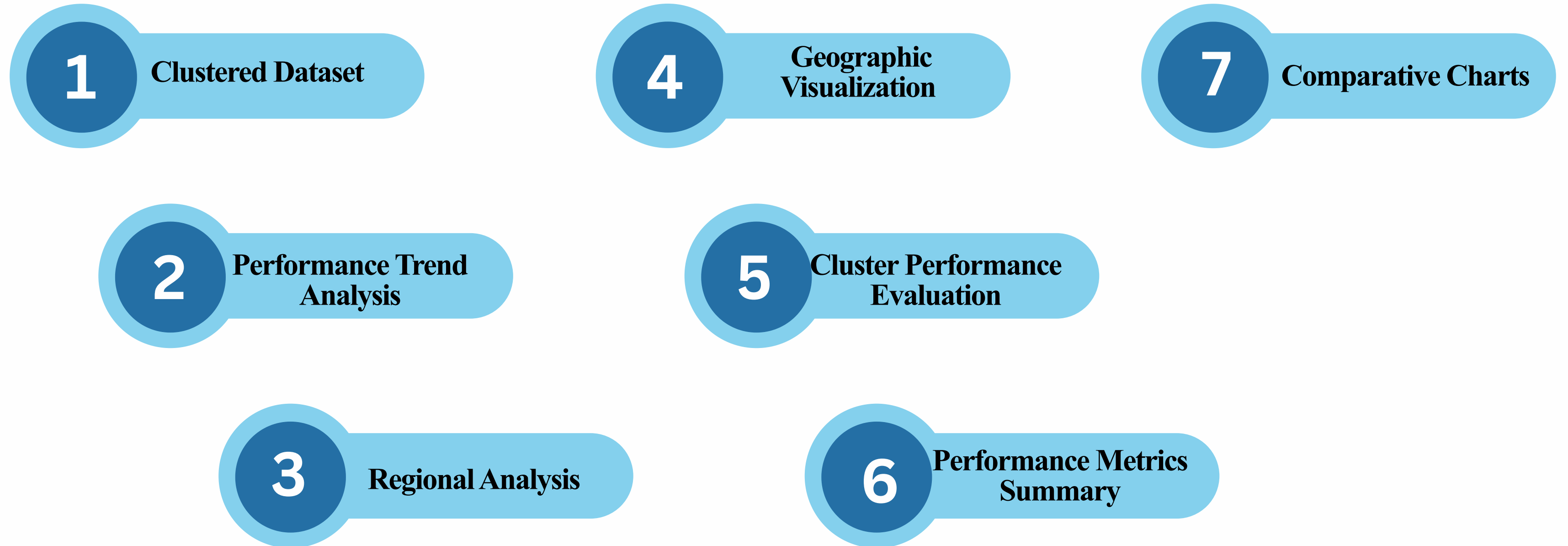
OBJECTIVES:

In this milestone, I focused on analyzing ticket performance trends and geographic patterns to evaluate operational efficiency across different regions and support categories.

ACTIVITIES PERFORMED:

- ✓ Resolution Time Trend Analysis
- ✓ Regional Performance Analysis
- ✓ Queue Performance Comparison
- ✓ High & Urgent Ticket Analysis
- ✓ Geographic Ticket Concentration Analysis
- ✓ Geographic Issue Category Visualization
- ✓ Cluster Size vs Performance Score Analysis
- ✓ Regional Performance Metrics & Comparative Charts

WORKFLOW

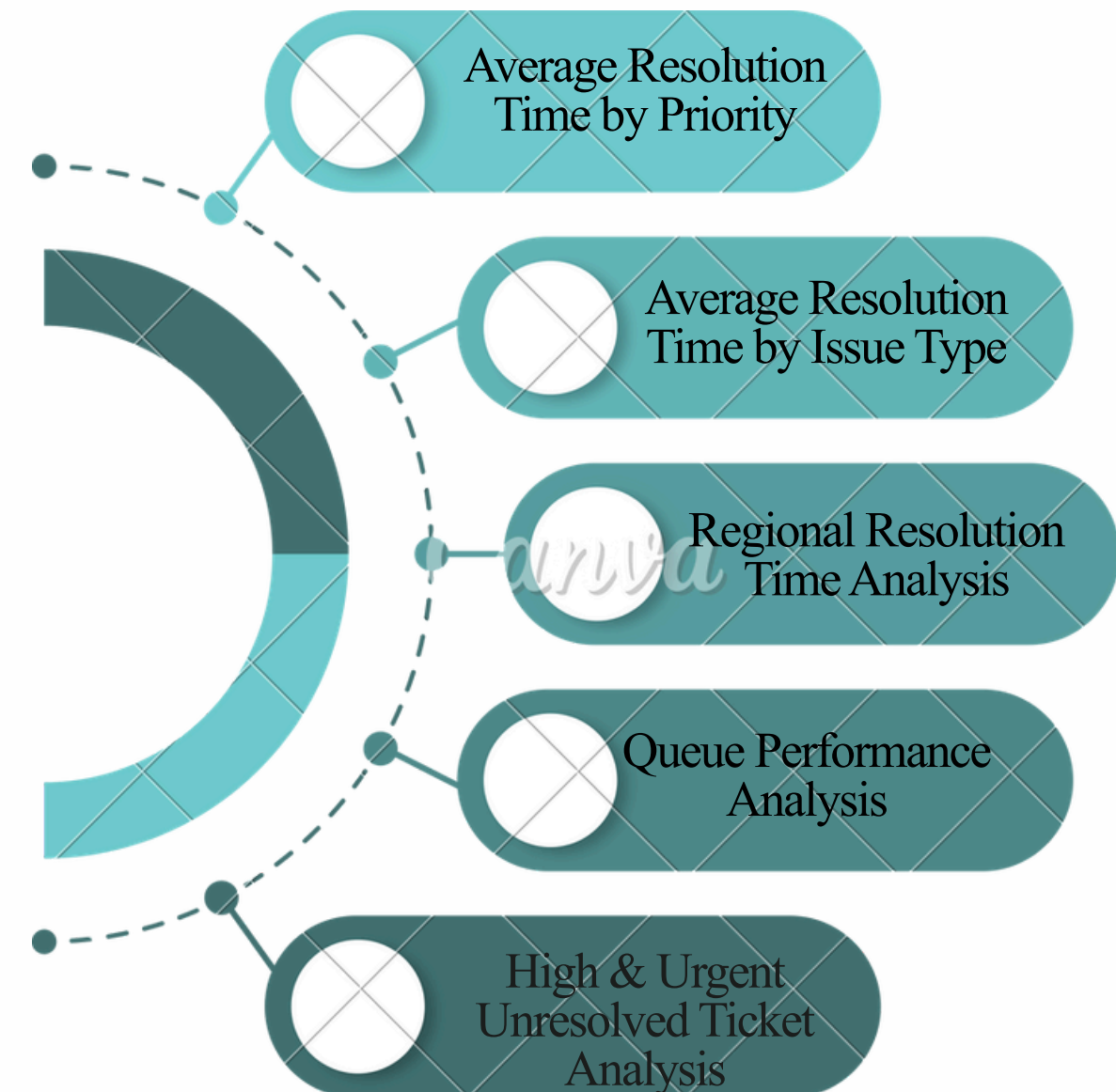


MODULE 5: PERFORMANCE TREND ANALYSIS

OBJECTIVE

To analyze ticket resolution performance across priorities, issue types, regions, and support queues, and identify critical unresolved support issues.

ACTIVITIES PERFORMED



AVERAGE RESOLUTION TIME ANALYSIS:

OBJECTIVE:



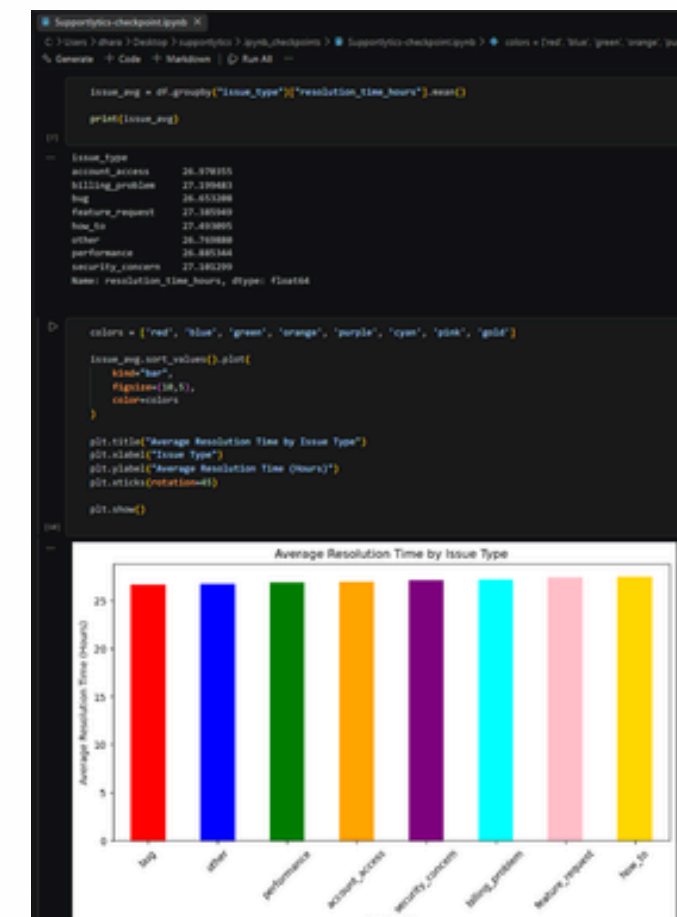
To identify which priority levels and issue types require longer resolution time.

KEY FINDINGS:

- Identified ticket categories with higher resolution time.
- Compared resolution efficiency across different ticket types.
- Highlighted areas that require process improvement

METHODOLOGY:

- Grouped tickets by Priority.
- Grouped tickets by Issue Type.
- Calculated average resolution time
- Visualized the results using bar charts



REGIONAL & QUEUE PERFORMANCE ANALYSIS

OBJECTIVE



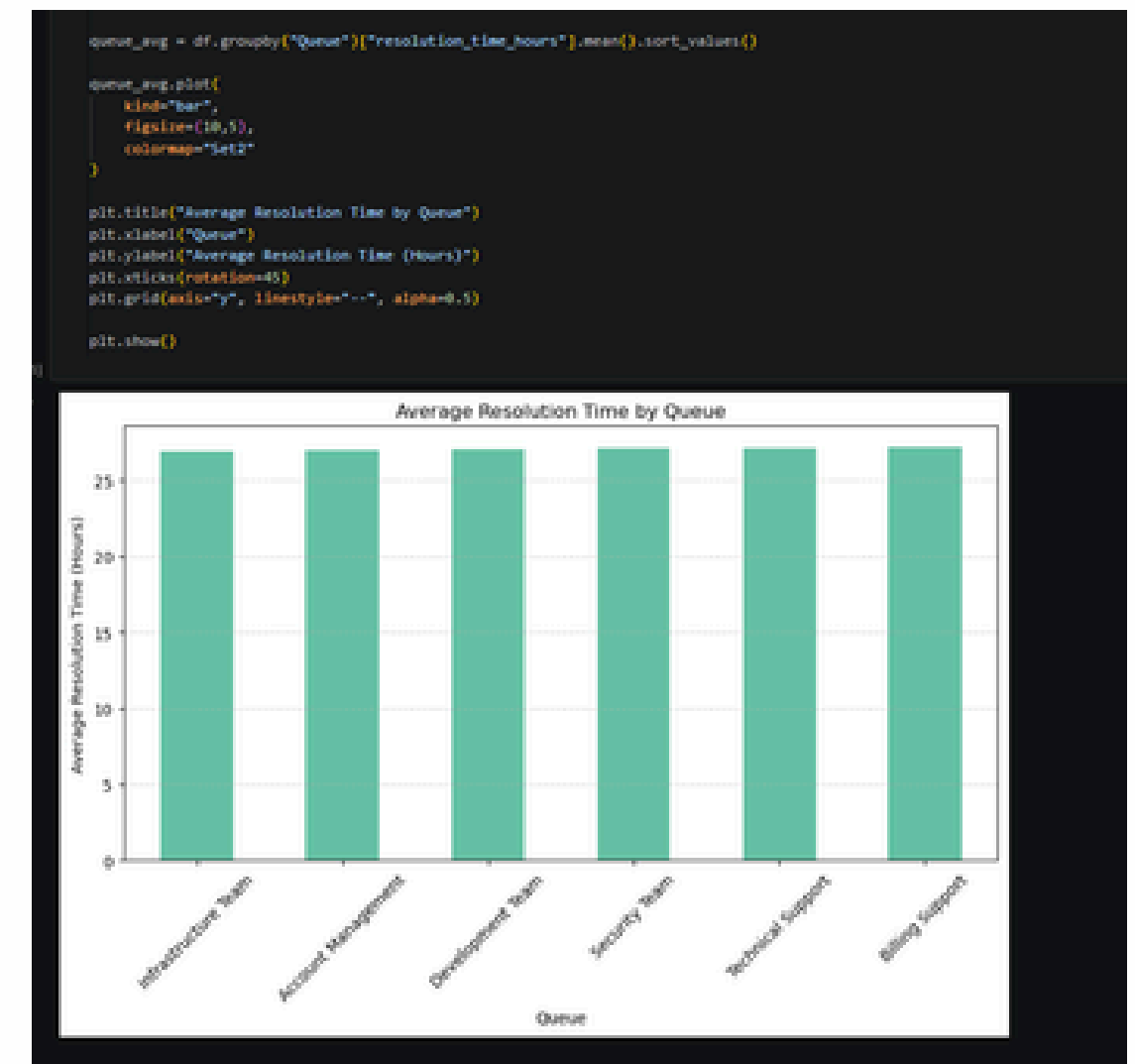
To compare ticket resolution efficiency across different regions and support queues.

KEY FINDINGS

- Identified regions that resolve tickets faster.
- Compared workload and efficiency across support queues.
- Evaluated regional operational performance.

METHODOLOGY

- Grouped tickets by Region.
- Grouped tickets by Queue.
- Calculated average resolution time
- Created comparative visualizations.



HIGH & URGENT UNRESOLVED ISSUES:

OBJECTIVE:



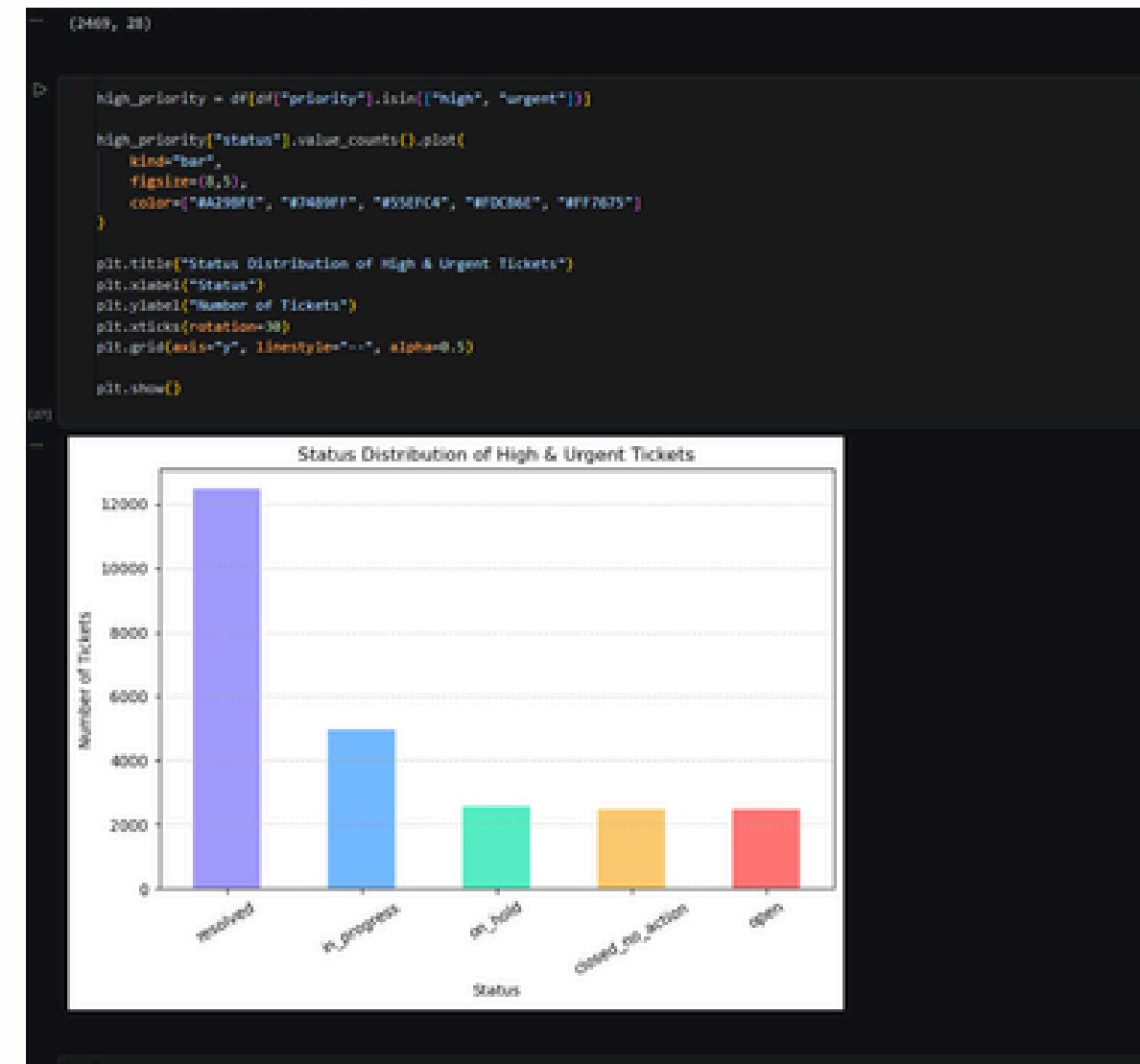
To identify critical unresolved support tickets requiring immediate attention.

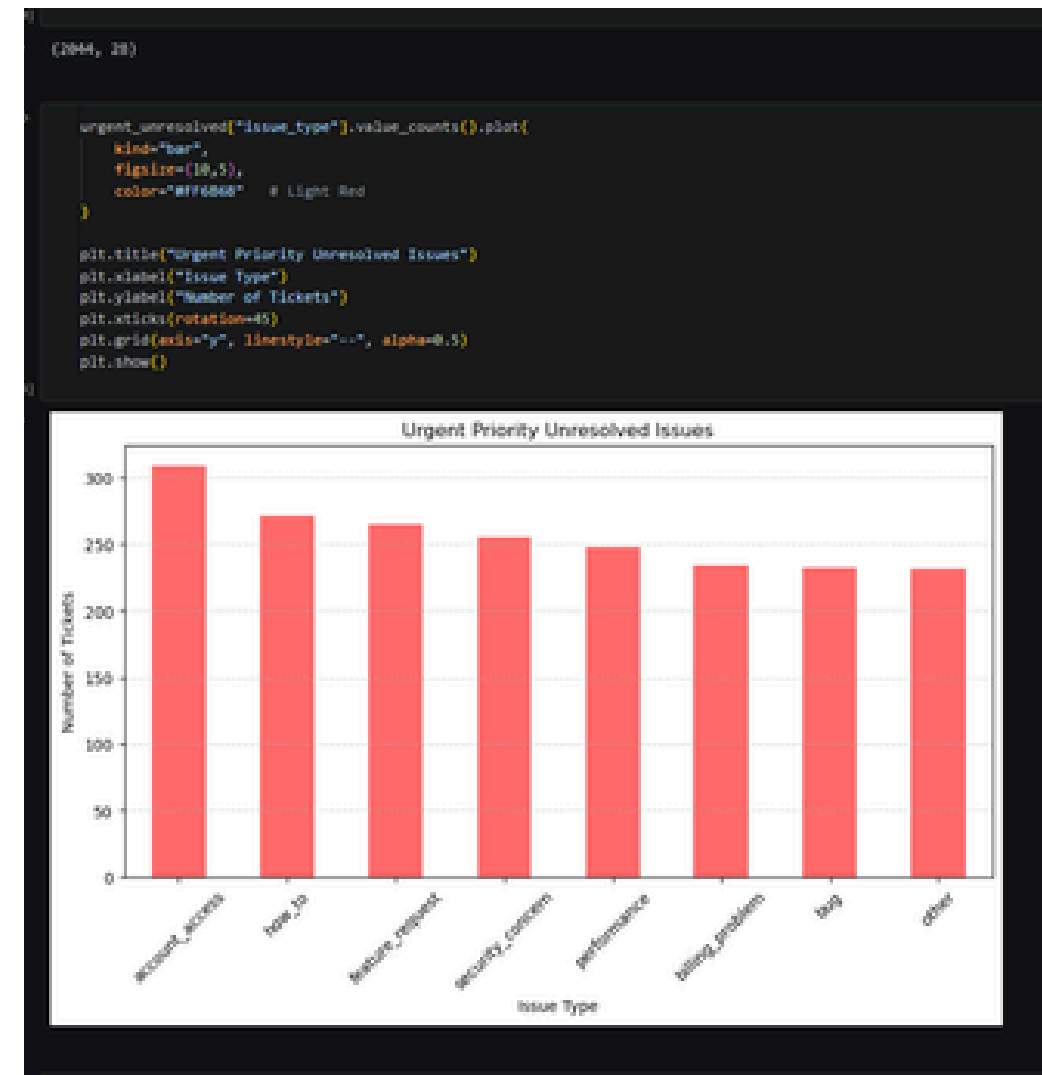
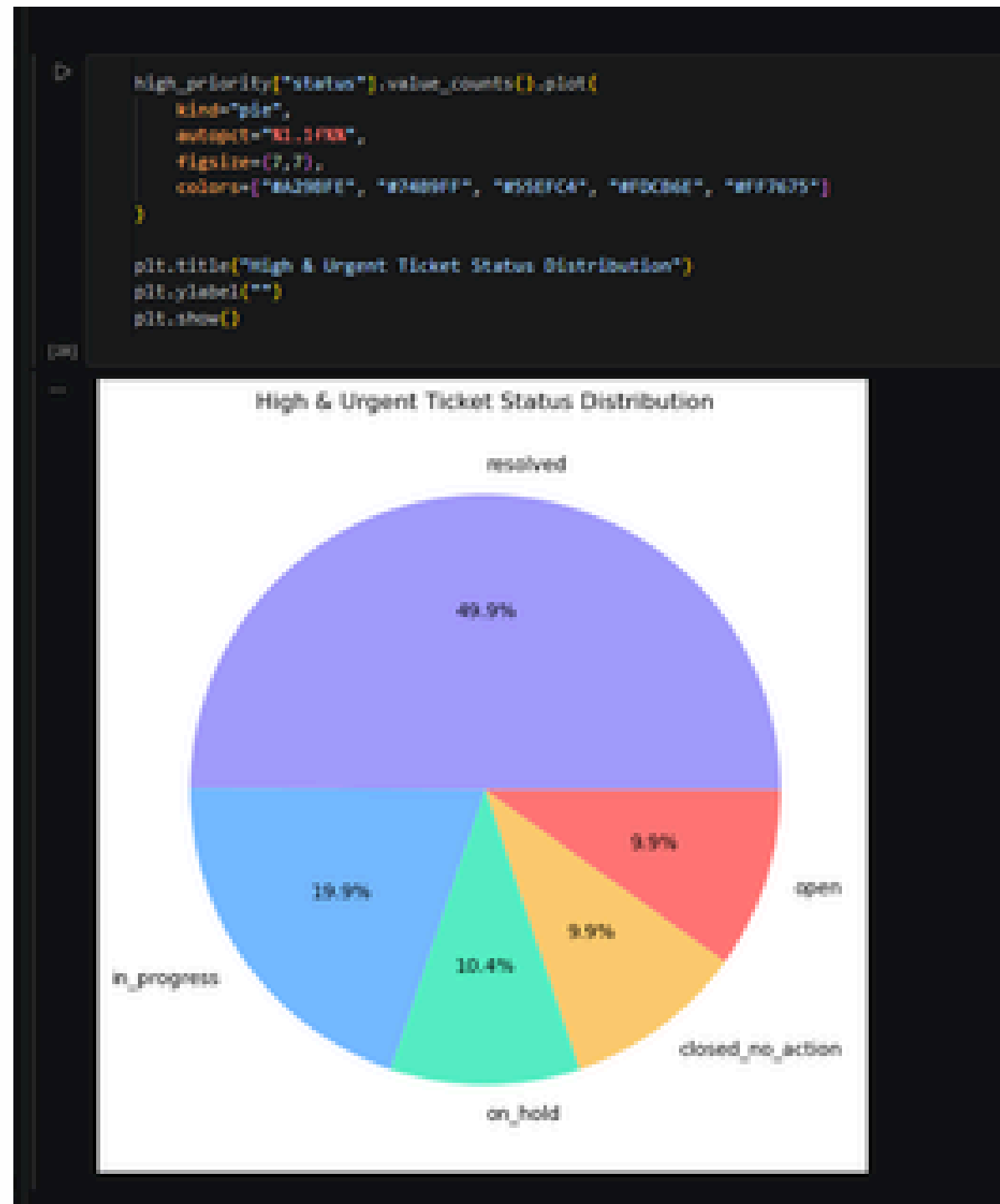
METHODOLOGY:

- Filtered High priority unresolved tickets.
- Filtered Urgent priority unresolved tickets.
- Counted unresolved tickets by Issue Type.
- Created separate visualizations for High and Urgent tickets

KEY FINDINGS:

- Identified major unresolved issue categories.
- Highlighted operational bottlenecks.
- Prioritized critical support requests.





Urgent priority
unresolved tickets

High priority
unresolved tickets



MODULE 6: GEOGRAPHIC & CATEGORY-LEVEL INSIGHTS

OBJECTIVE:

To analyze regional ticket distribution and evaluate geographic and cluster-level performance across different support regions.

ACTIVITIES PERFORMED:

- Regional Ticket Concentration Analysis
- Geographic Ticket Mapping
- Issue Category Geographic Analysis
- Cluster Size vs Performance Score Analysis
- Regional Performance Metrics
- Comparative Regional Charts

REGIONAL TICKET CONCENTRATION:

OBJECTIVE:



To identify regions generating the highest number of support tickets.

KEY FINDINGS:

- Identified regions with the highest support workload.
- Visualized regional ticket concentration patterns.
- Supported geographic resource planning.

METHODOLOGY:

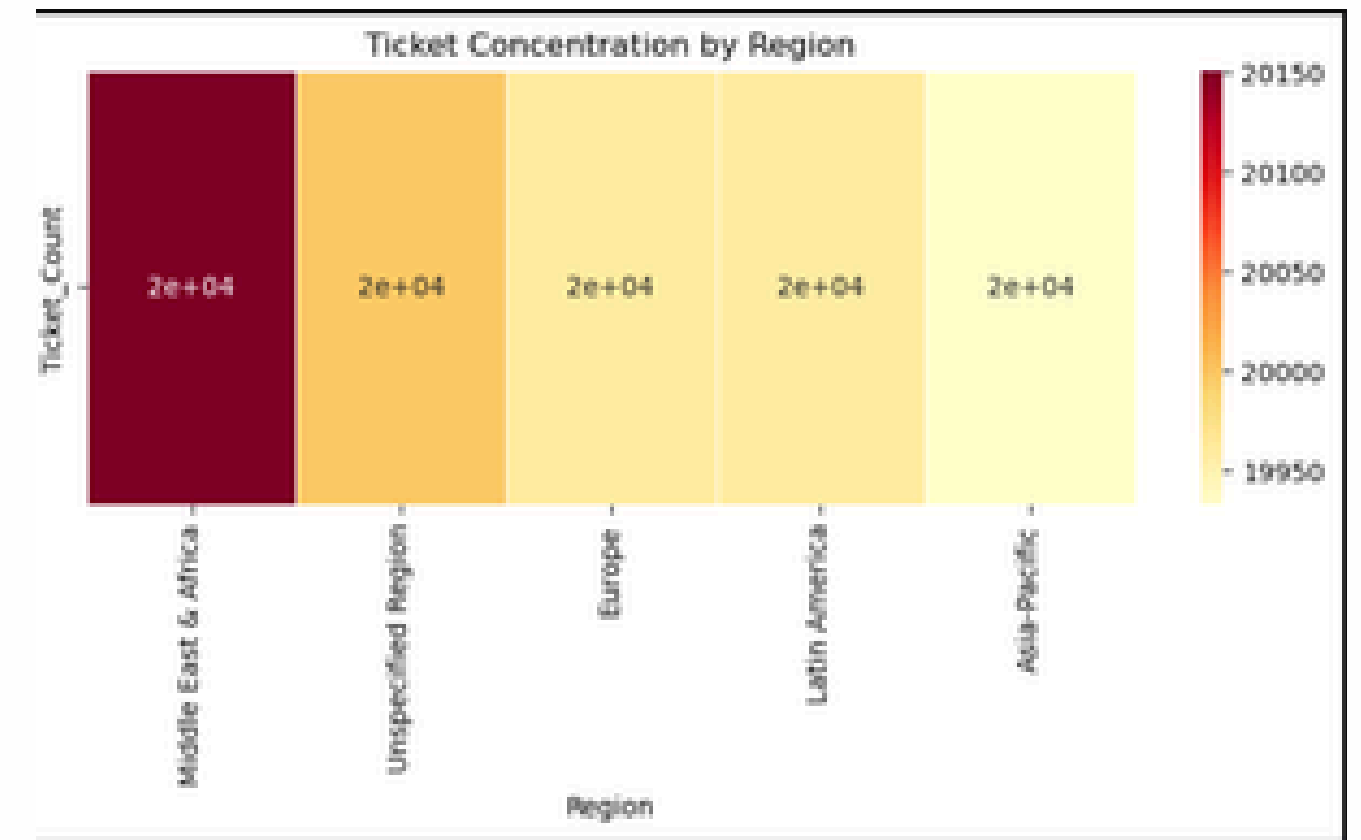
- Standardized region names.
- Counted ticket frequency by region.
- Created a heatmap of ticket concentration.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(9,3))

sns.heatmap(
    region_counts.set_index('Region').T,
    annot=True,
    cmap='YlOrRd',
    linewidths=0.5
)

plt.title('Ticket Concentration by Region')
plt.show()
```



GEOGRAPHIC DISTRIBUTION OF ISSUE CATEGORIES

OBJECTIVE:



To visualize support ticket distribution geographically using latitude and longitude.

KEY FINDINGS:

- Visualized global ticket distribution.
- Identified geographic concentration of issue categories.
- Improved understanding of regional support patterns.

METHODOLOGY:

- Assigned latitude and longitude coordinates for each region.
- Created a global geographic visualization.
- Mapped issue categories across different regions.

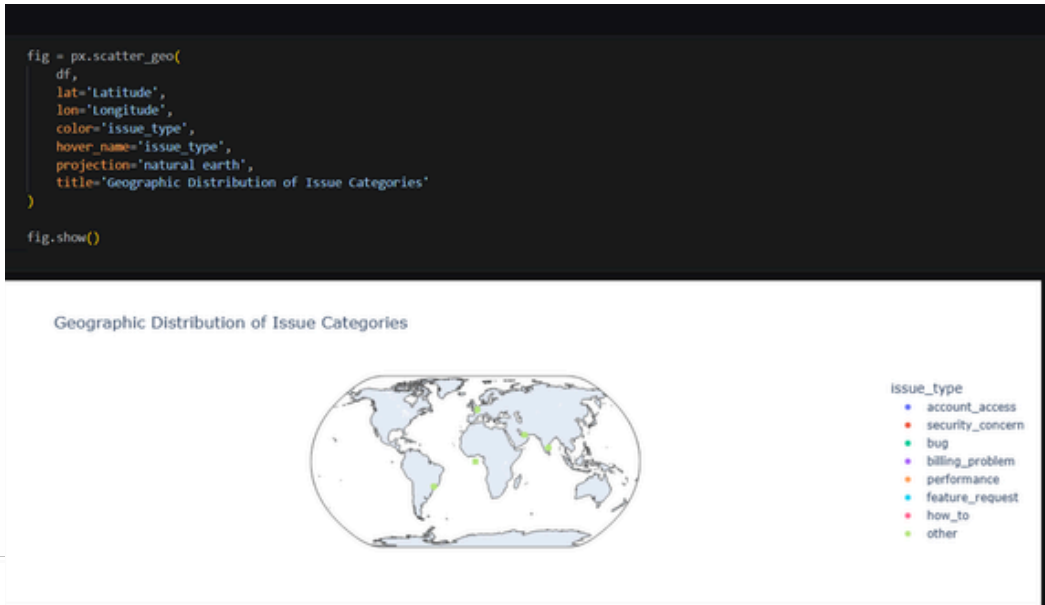
```
region_summary = df.groupby("region").size().reset_index(name="Ticket_Count")

region_summary["Latitude"] = region_summary["region"].map(lambda x: coordinates[x][0])
region_summary["Longitude"] = region_summary["region"].map(lambda x: coordinates[x][1])

fig = px.scatter_geo(
    region_summary,
    lat="Latitude",
    lon="Longitude",
    size="Ticket_Count",
    color="region",
    hover_name="region",
    projection="natural earth",
    title="Global Ticket Concentration by Region"
)

fig.show()
```

Global Ticket Concentration by Region



Geographic Distribution of Issue Categories

CLUSTER PERFORMANCE ANALYSIS

OBJECTIVE:



To evaluate the relationship between cluster size and performance score.

KEY FINDINGS:

- Compared cluster size with customer performance.
- Identified high-volume clusters with lower performance.
- Supported cluster-level operational analysis.

METHODOLOGY:

- Used clusters generated from Module 4.
- Converted customer sentiment into a numerical Performance Score
- Calculated cluster size and average performance score.
- Created a scatter plot visualization

```
import matplotlib.pyplot as plt
import seaborn as sns

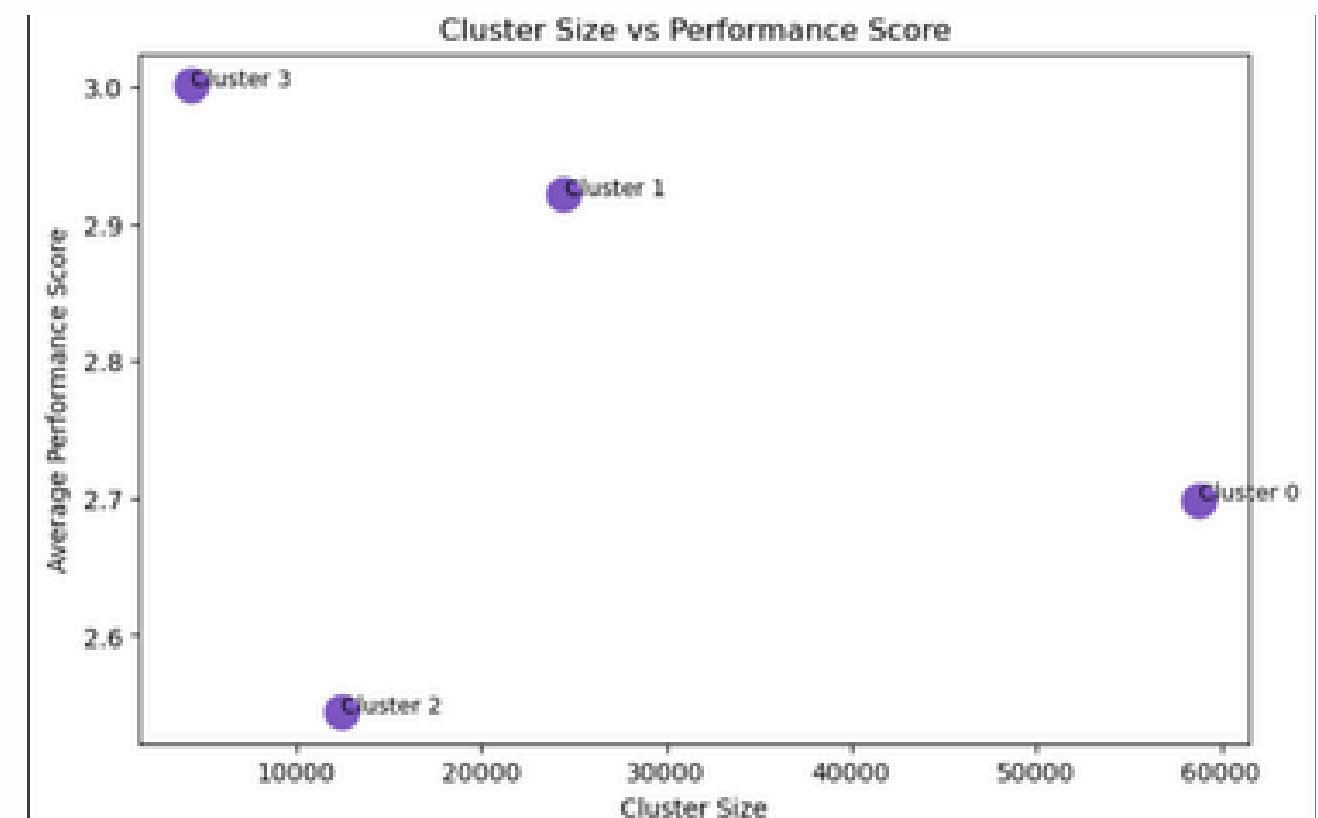
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=cluster_performance,
    x='Cluster_Size',
    y='Avg_Performance_Score',
    s=250,
    color='#7E57C2'
)

for i, row in cluster_performance.iterrows():
    plt.text(
        row['Cluster_Size'] + 20,
        row['Avg_Performance_Score'],
        f'Cluster {int(row["cluster"])}',
        fontsize=9
    )

plt.title('Cluster Size vs Performance Score')
plt.xlabel('Cluster Size')
plt.ylabel('Average Performance Score')

plt.show()
```



REGIONAL PERFORMANCE METRICS

OBJECTIVE:



To summarize overall support performance across different geographic regions.

METHODOLOGY:

Calculated:

- Average Resolution Time
- Average Performance Score
- Average CSAT Score
- Ticket Count

Created comparative regional visualizations.

KEY FINDINGS:

- Compared operational efficiency across regions.
- Evaluated regional customer satisfaction.
- Generated performance metrics summary.

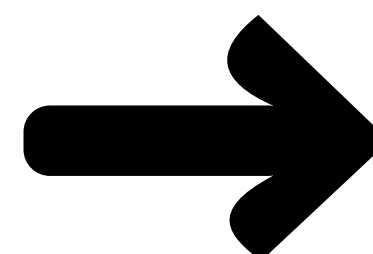
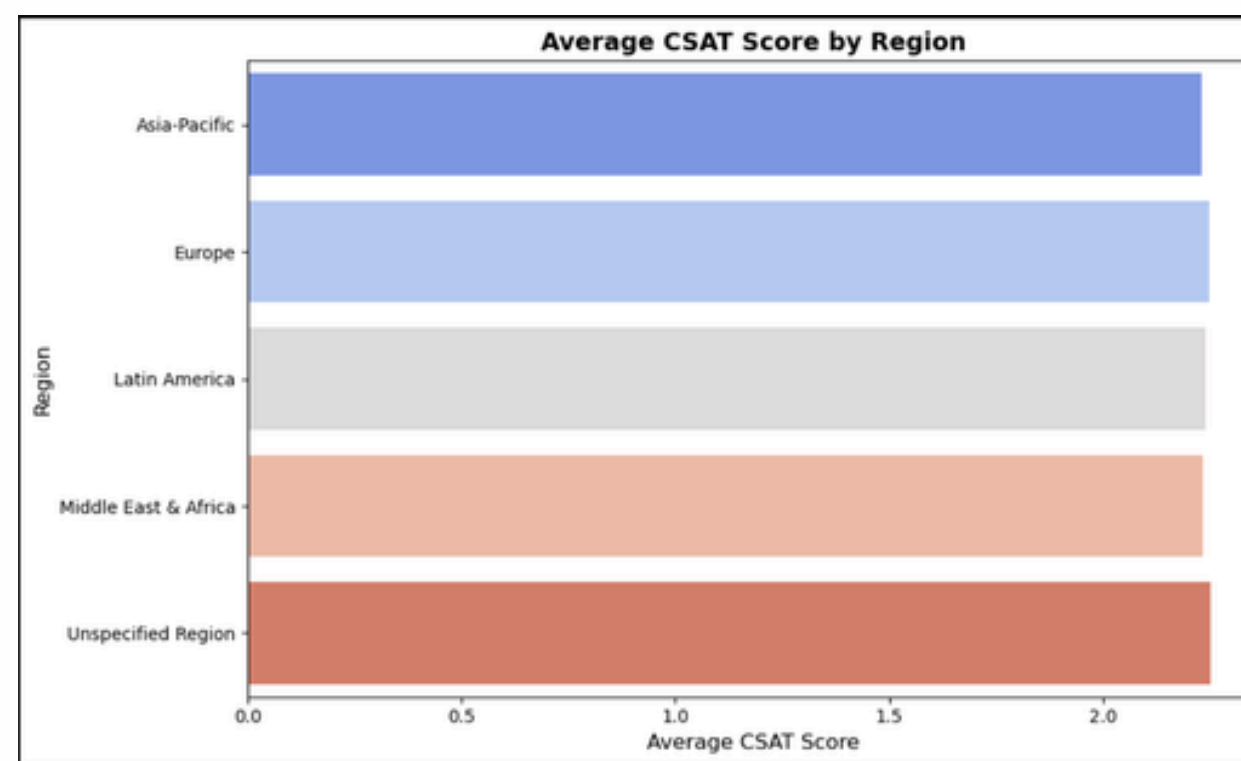
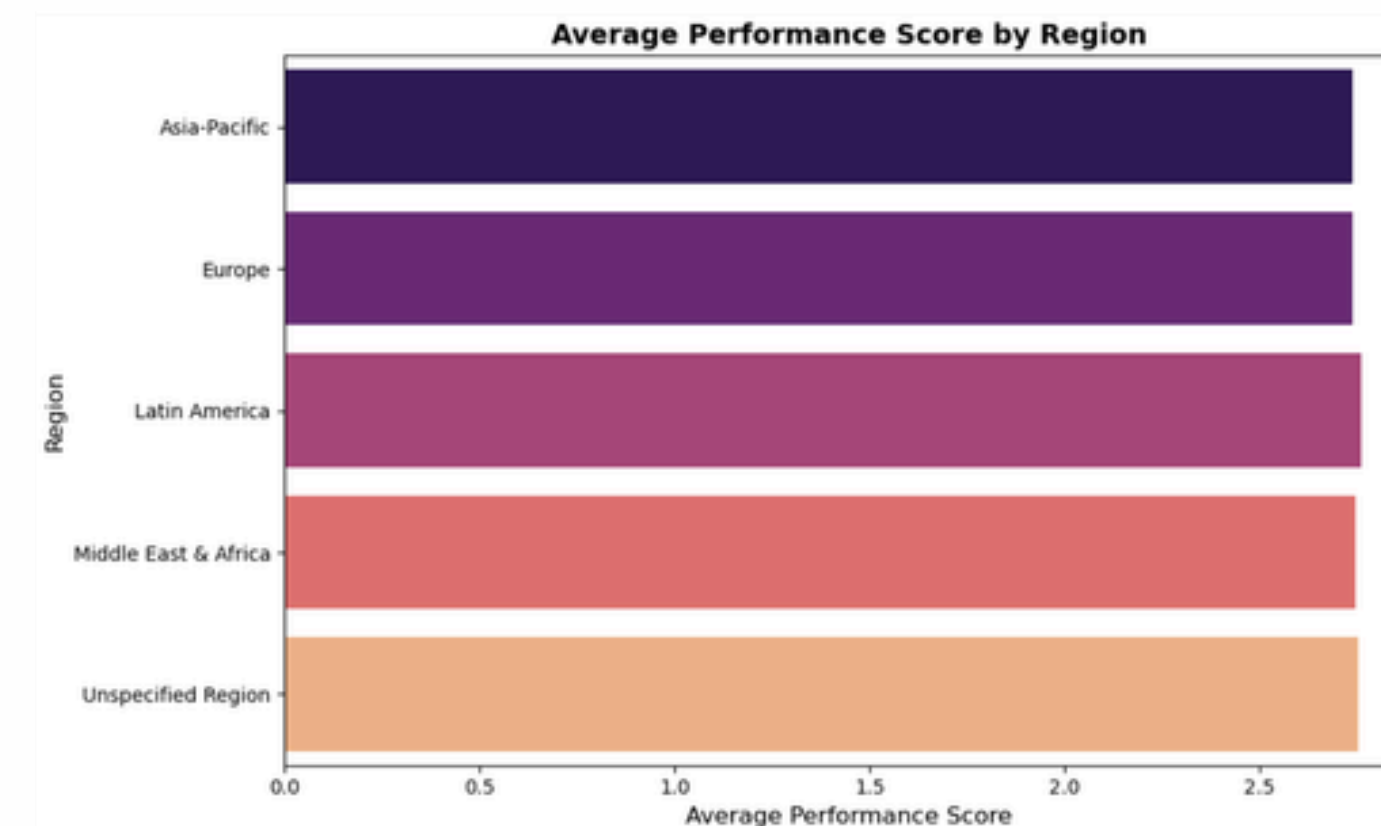
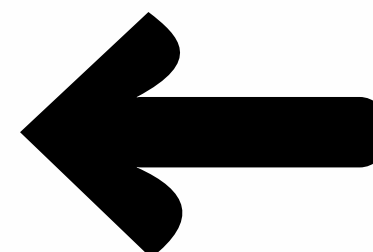
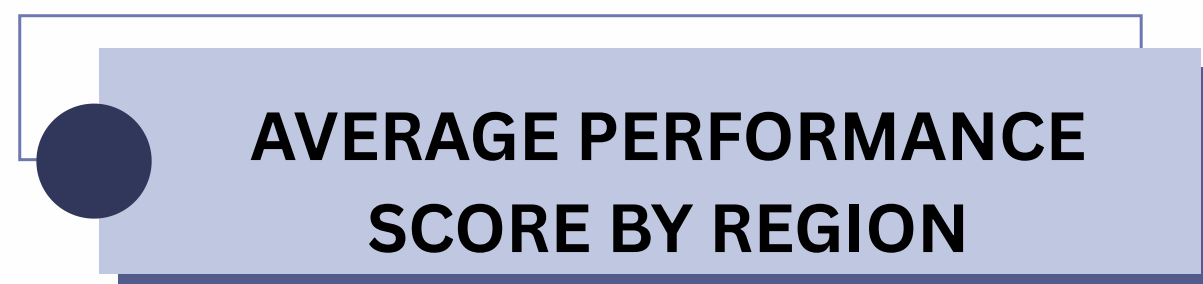
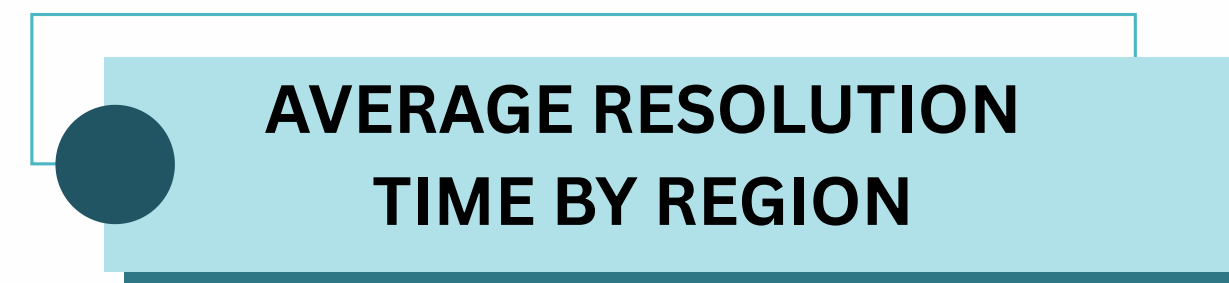
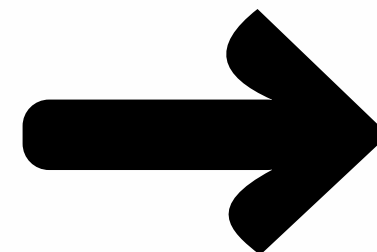
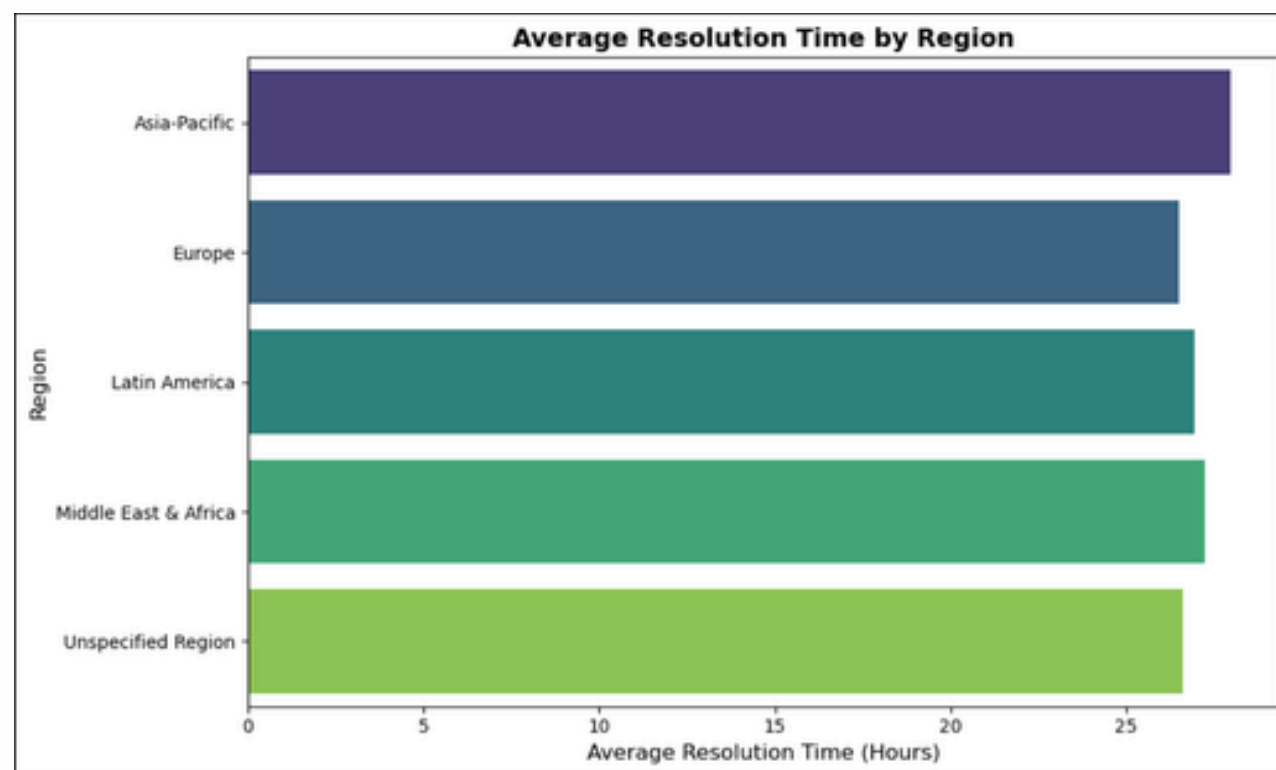
```
[43]: performance_summary = df.groupby('region').agg(
    Avg_Resolution_Times('resolution_time_hours', 'mean'),
    Avg_Performance_Scores('Performance_Score', 'mean'),
    Avg_CSATs('csat_score', 'mean'),
    Ticket_Counts('region', 'count')
).reset_index()

performance_summary
```

```
[43]:
```

	region	Avg_Resolution_Time	Avg_Performance_Score	Avg_CSAT	Ticket_Count
0	Asia-Pacific	27.965975	2.739427	2.232328	19933
1	Europe	26.515893	2.737275	2.250100	19960
2	Latin America	26.959489	2.760309	2.240844	19959
3	Middle East & Africa	27.238882	2.746067	2.234827	20151
4	Unspecified Region	26.613777	2.751763	2.251688	19997

```
[ ]:
```

KEY ACHIEVEMENTS:

MILESTONE DELIVERABLES:

- ✓ Resolution Time Analysis Completed
- ✓ Regional & Queue Performance Evaluation Completed
- ✓ High & Urgent Ticket Analysis Completed
- ✓ Geographic Ticket Visualization Completed
- ✓ Cluster Performance Evaluation Completed
- ✓ Performance Metrics Summary Generated
- ✓ Comparative Regional Charts Created

CONCLUSION:

» Milestone 3 successfully extended the Supportlytics project by analyzing performance trends and geographic support patterns. The generated insights support data-driven decision-making, improve ticket resolution efficiency, and help optimize IT support operations across different regions.

MILESTONE - 4

MODULE 7

OBJECTIVE:



The objective of Milestone 4 is to create a complete Power BI dashboard that combines the insights generated from the previous milestones into a single interactive business intelligence solution.

DASHBOARD FEATURES:

- Executive Overview
- Resolution Time Analysis
- Ticket Categories & Distribution
- Geographic Insights
- Cluster Similarity Analysis
- Regional Performance Dashboard
- Insights & Recommendations

Executive Overview Dashboard

Objective:



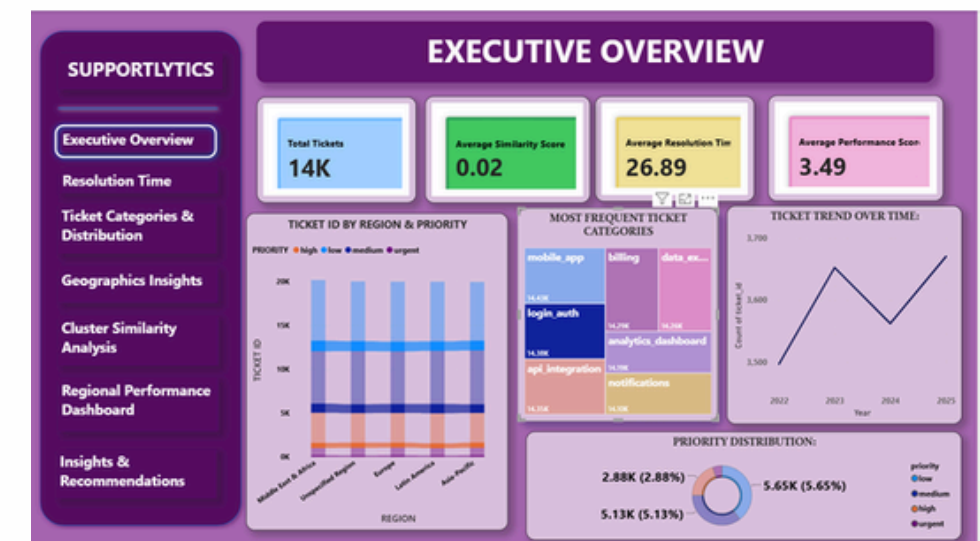
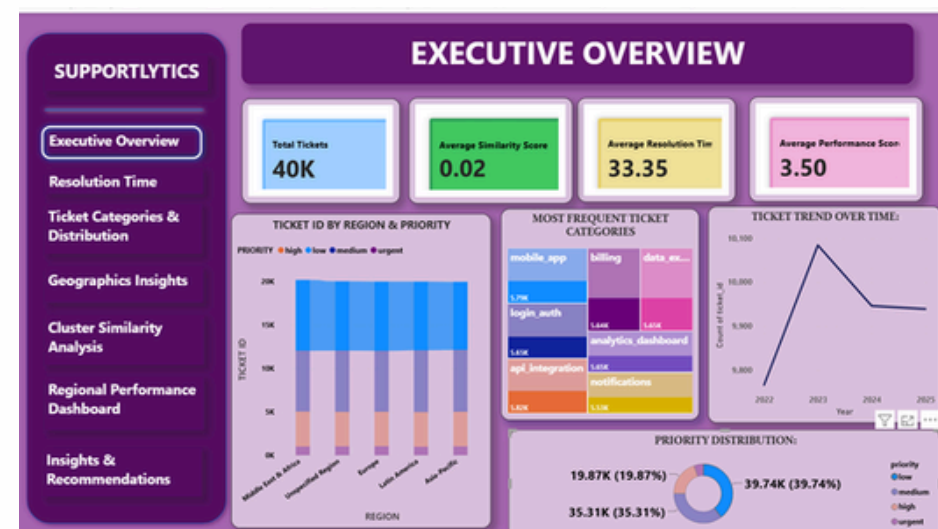
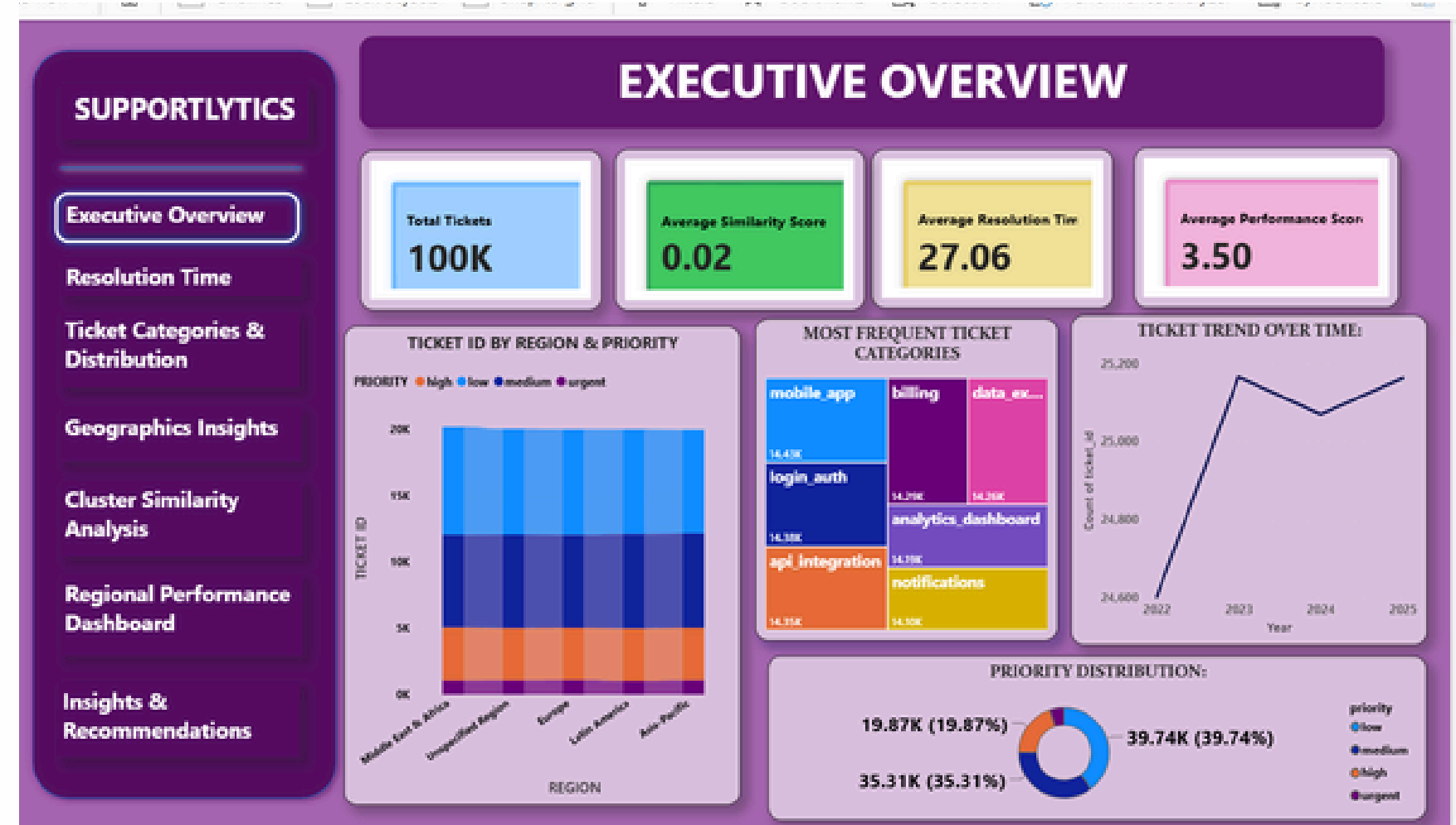
To provide a quick summary of the overall IT support ticket performance.

Visuals Used

- KPI Cards
- Treemap
- Donut Chart
- Regional Ranking Chart
- Ticket Trend Line Chart

Key Insights

- Total support tickets analyzed
- Average resolution time
- Average performance score
- Ticket category distribution
- Priority distribution
- Regional ticket ranking
- Ticket trend over time



Resolution Time Dashboard

Objective

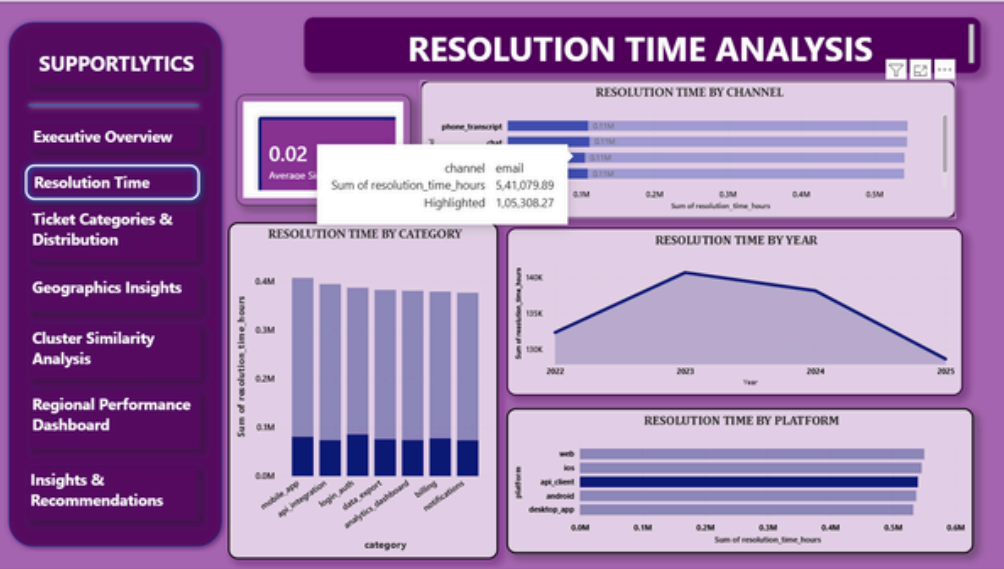
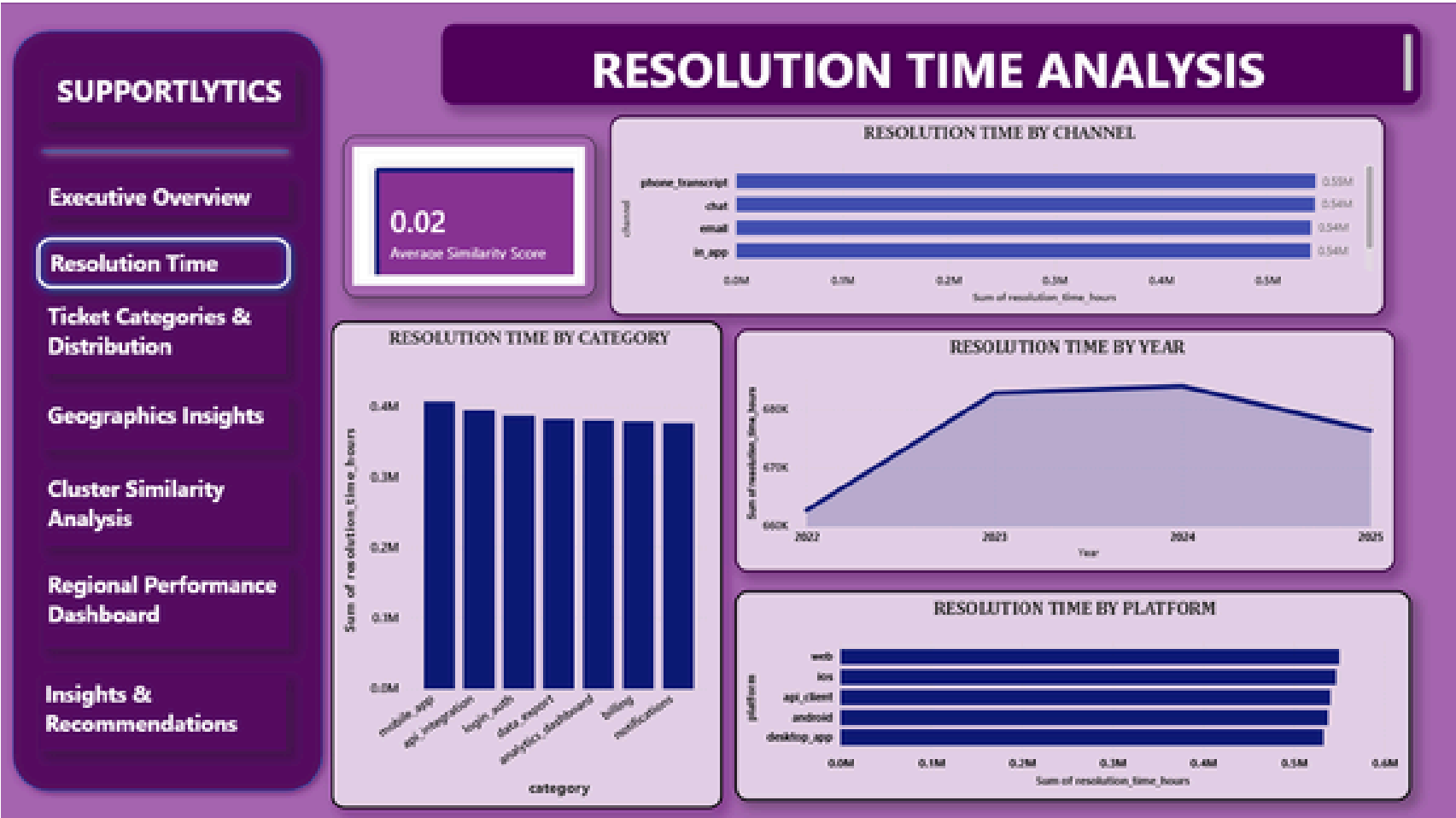
» To analyze how efficiently support tickets are resolved across priorities, issue types, regions, and support queues.

Visuals Used

- Gauge Chart
- Bar Chart
- Column Chart
- Heatmap
- Line Chart

Key Insights

- Resolution time by priority
- Resolution time by issue type
- Regional resolution comparison
- Queue performance analysis
- Resolution trend over time



Ticket Categories & Geographic Insights

Objective



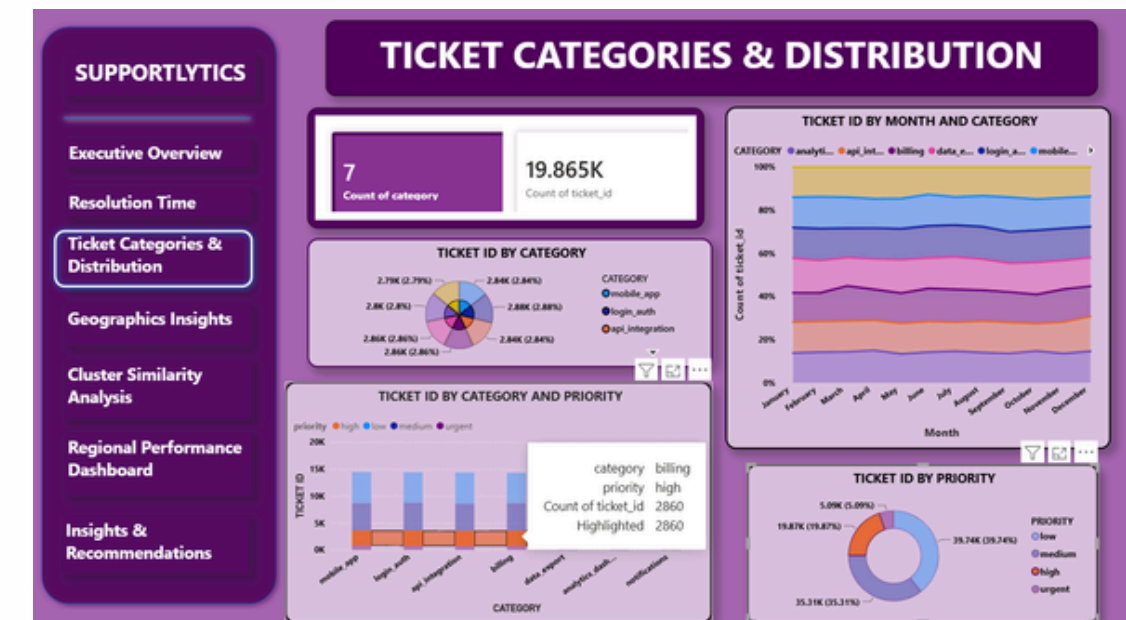
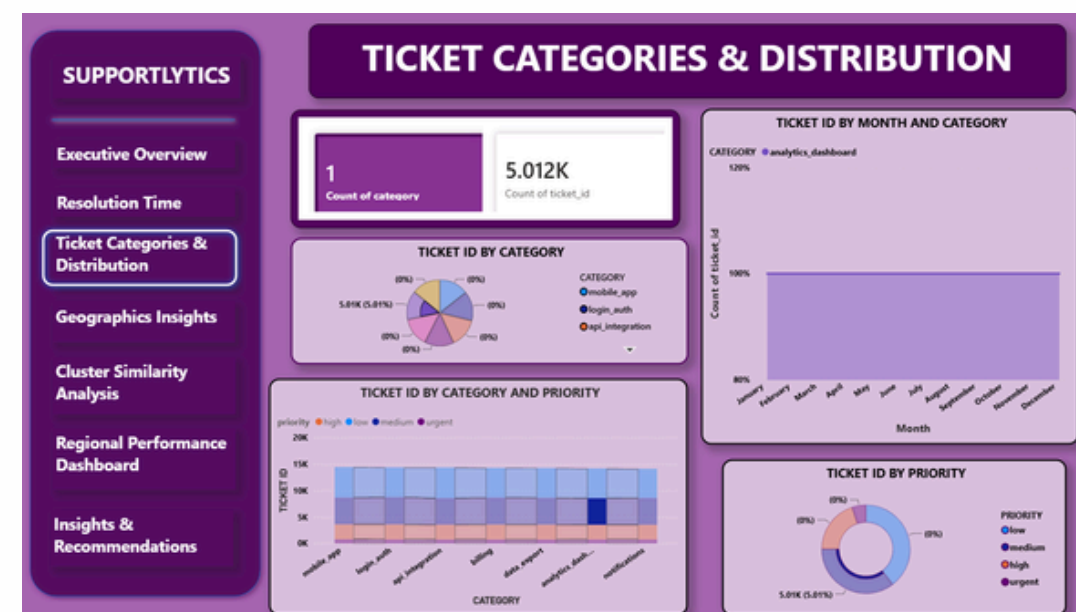
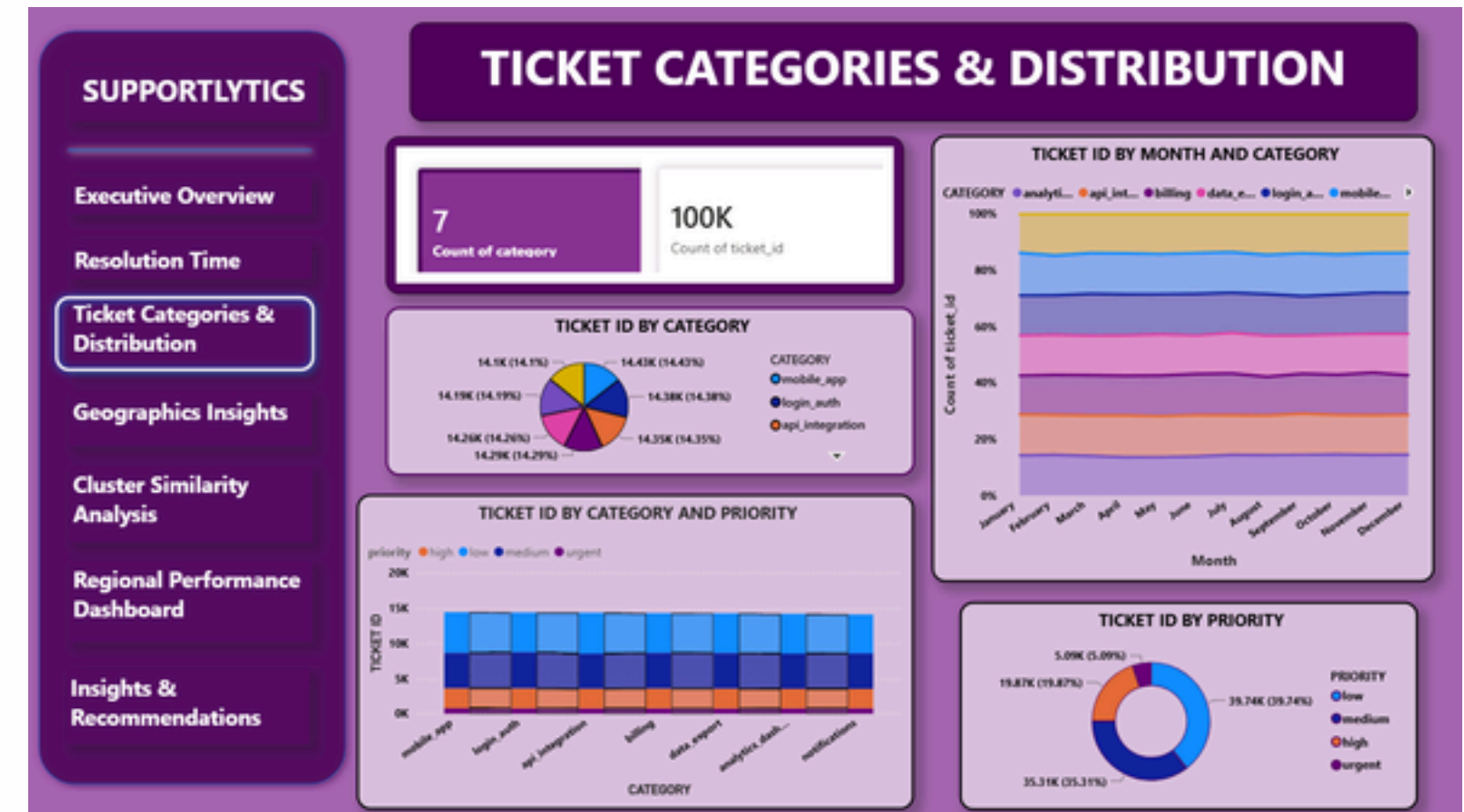
To understand ticket distribution across categories and geographic regions.

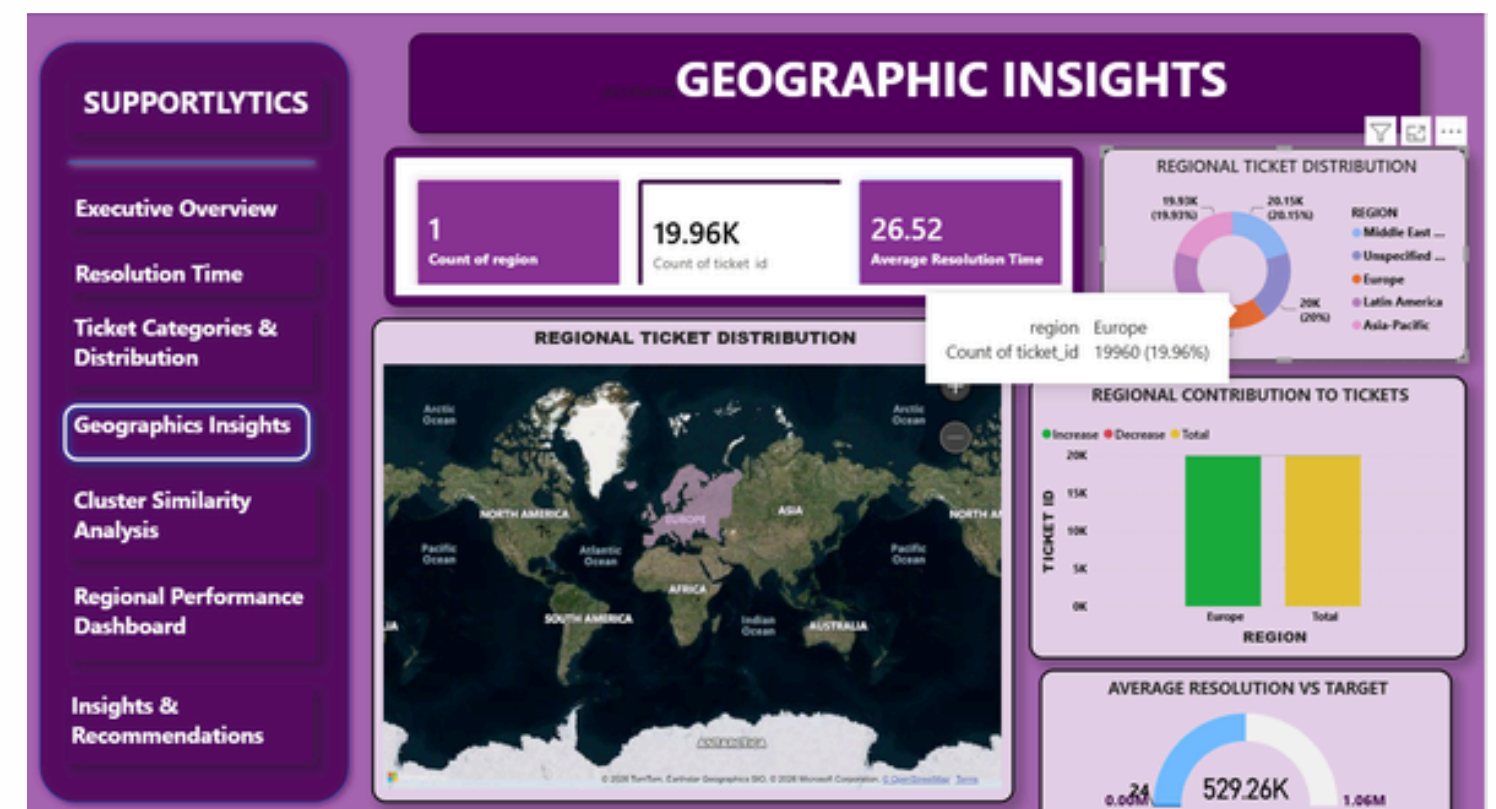
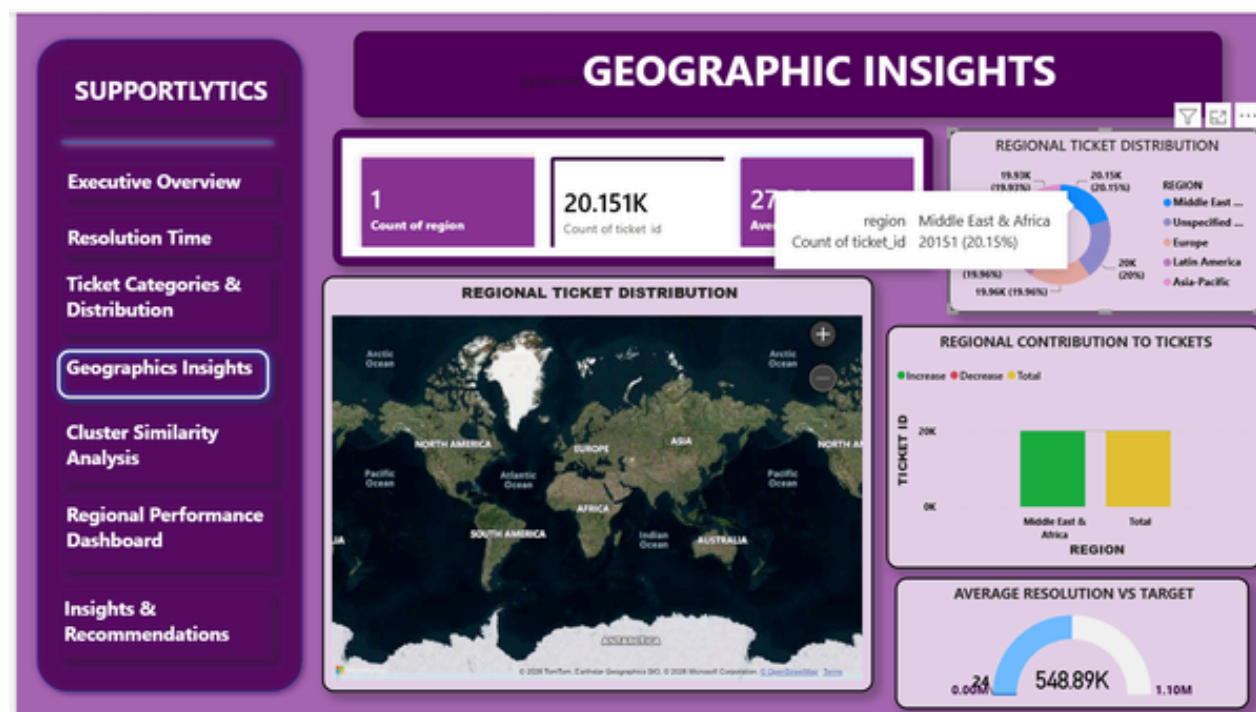
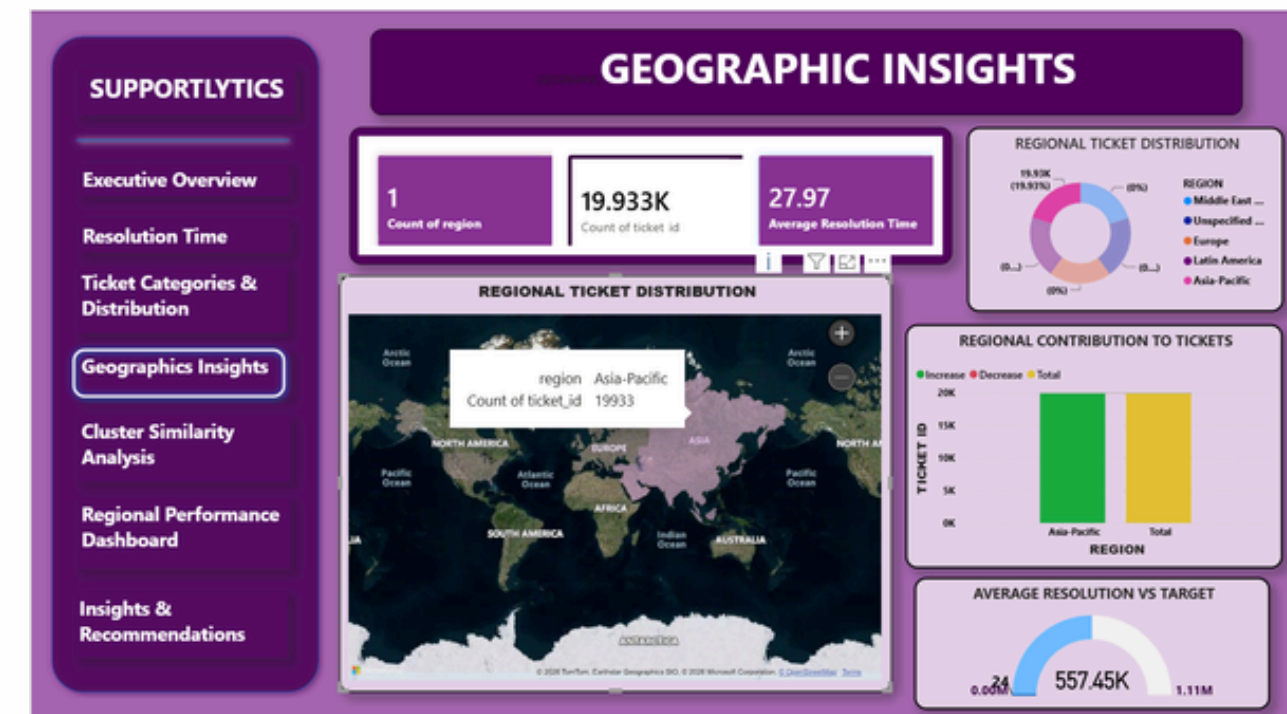
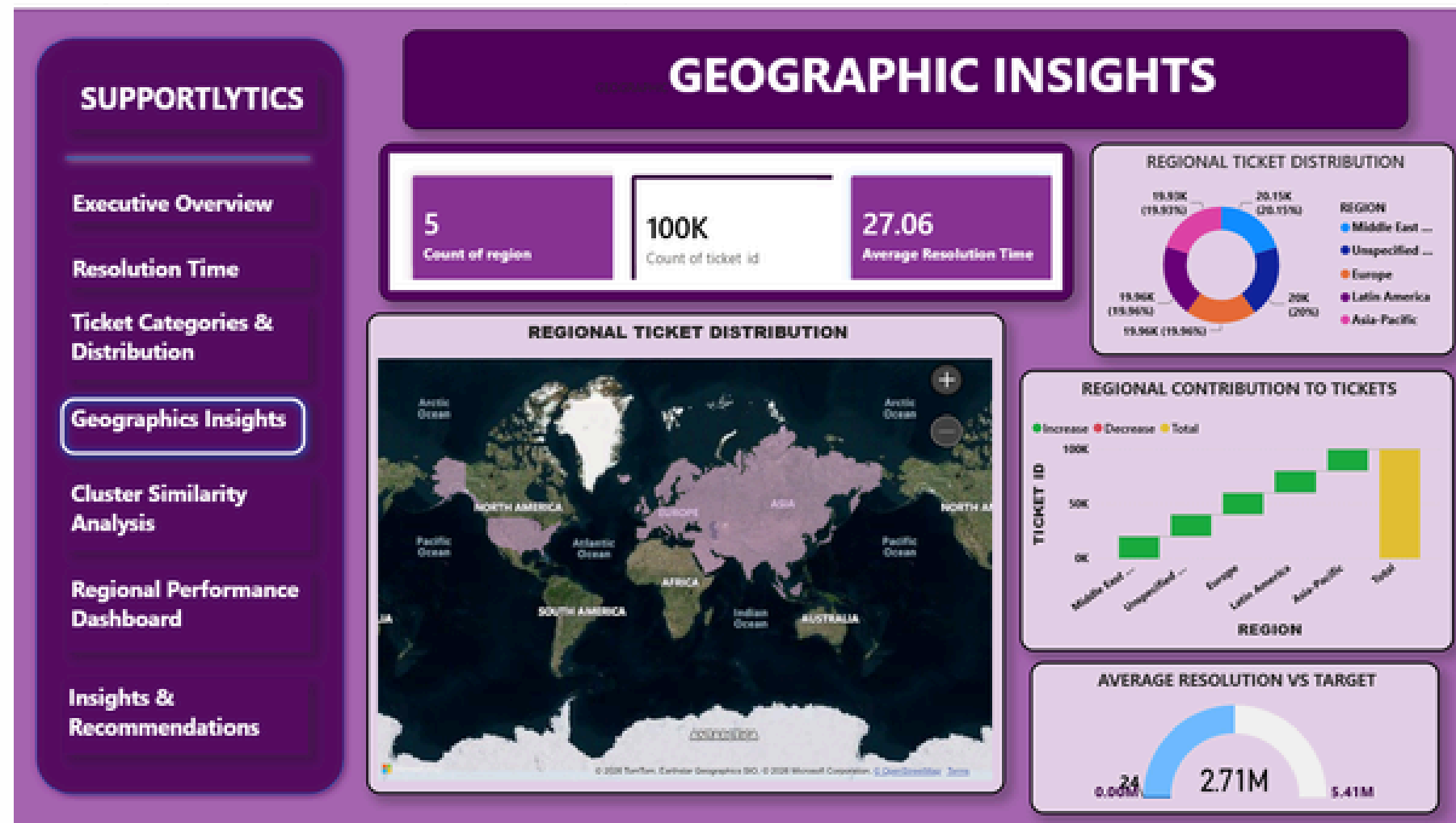
Visuals Used

- Treemap
- Stacked Bar Chart
- Funnel Chart
- Donut Chart
- Map Visual
- Bubble Map
- Heatmap Matrix

Key Insights

- Most frequent ticket categories
- Ticket status distribution
- Queue distribution
- Geographic ticket concentration
- Regional issue category patterns





Cluster Similarity & Regional Performance

Objective



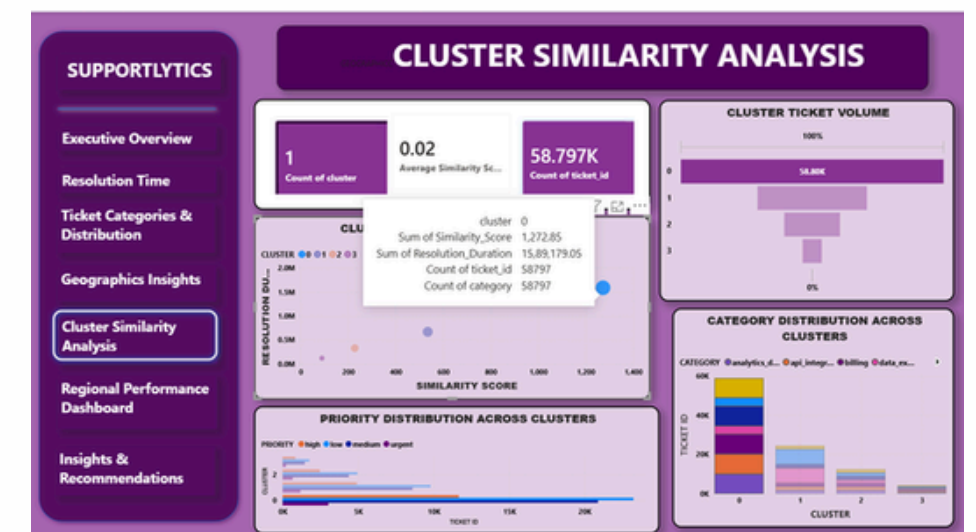
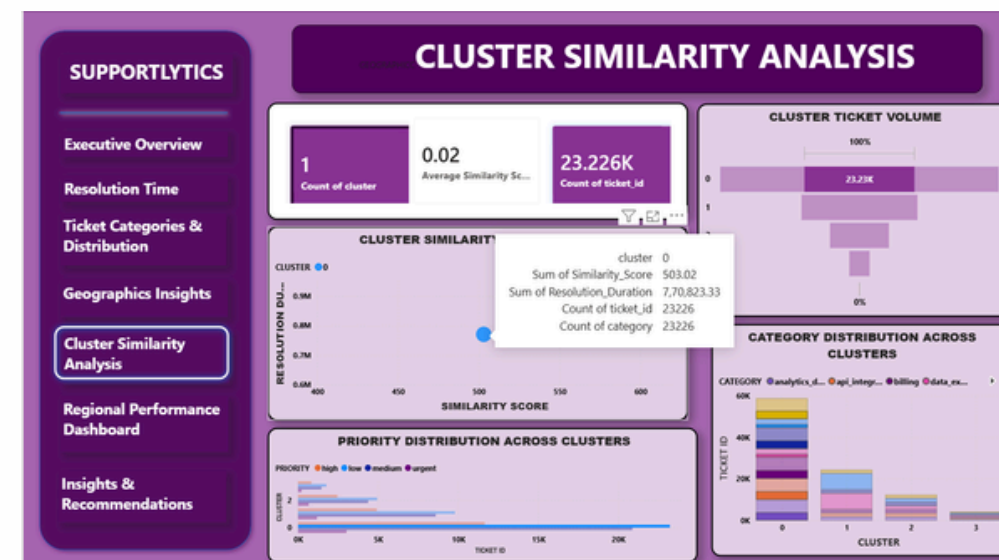
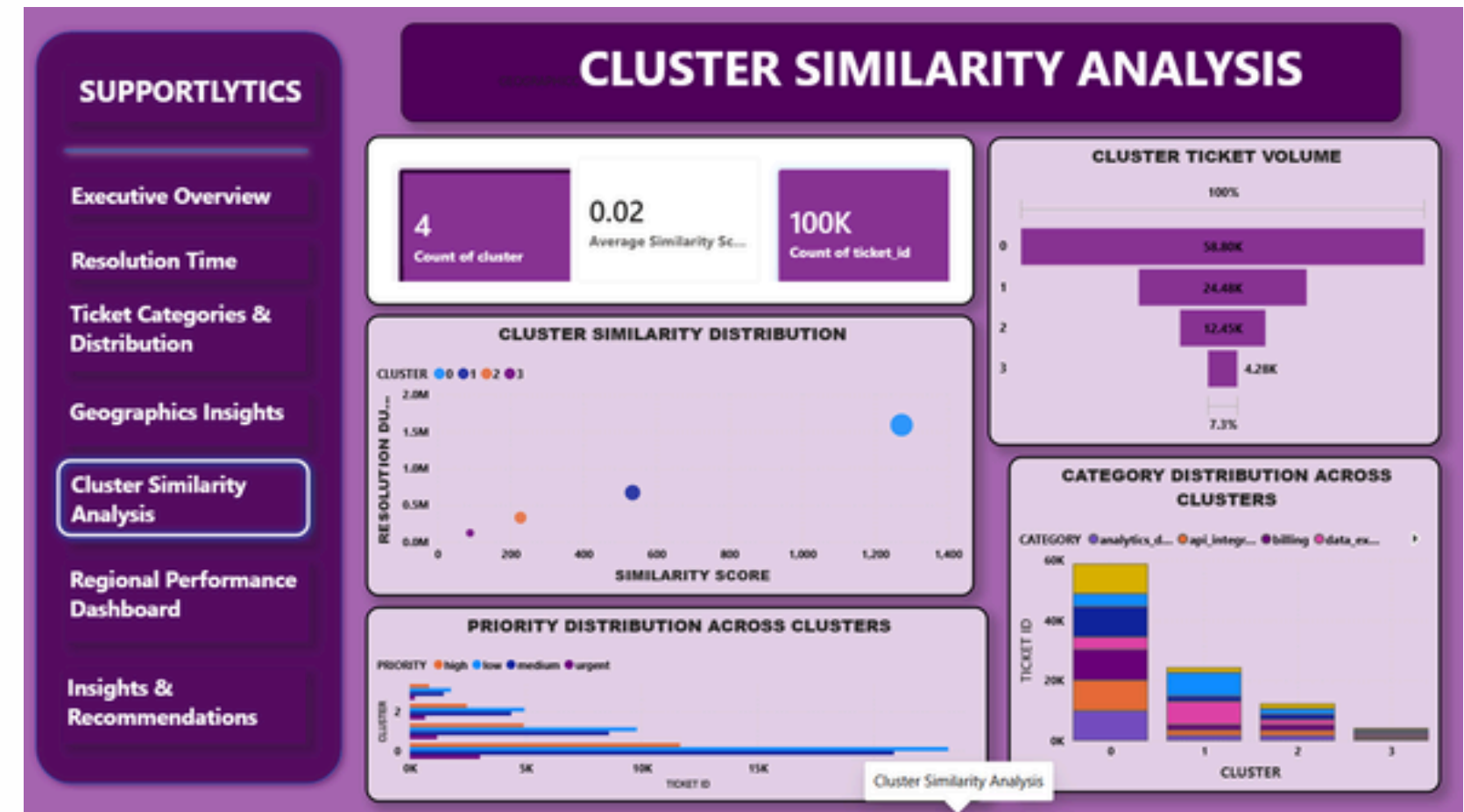
To evaluate ticket similarity patterns and compare operational performance across regions.

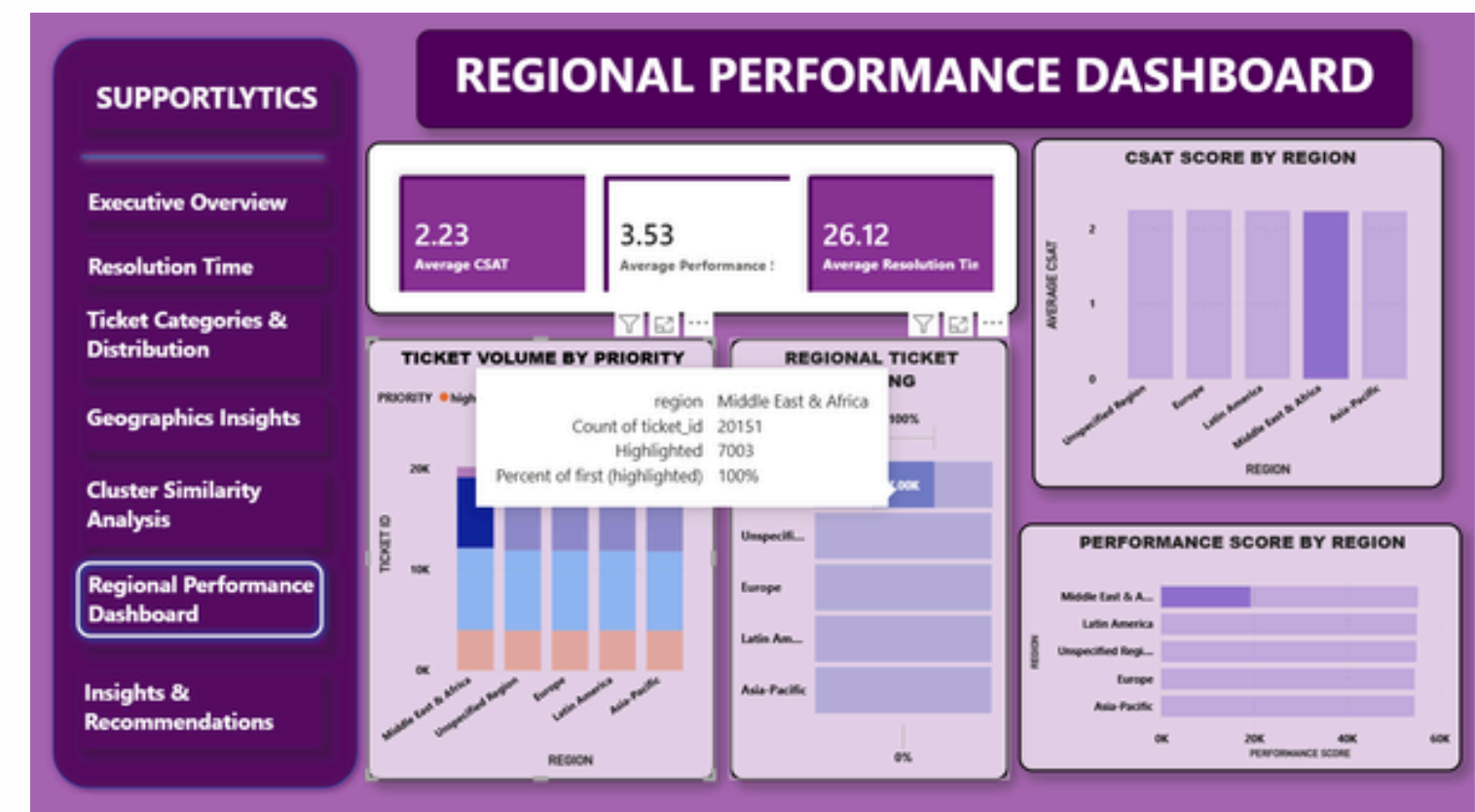
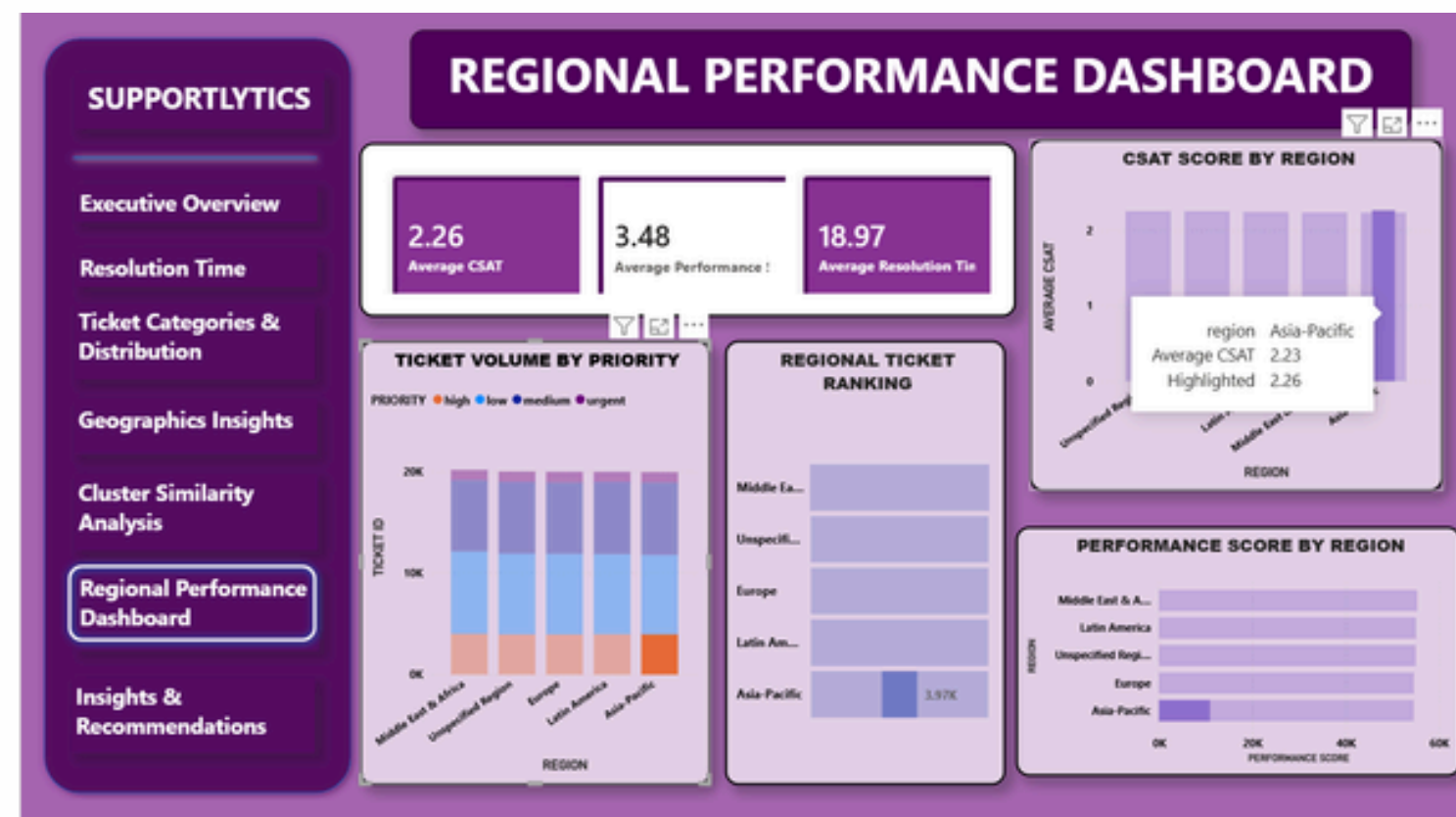
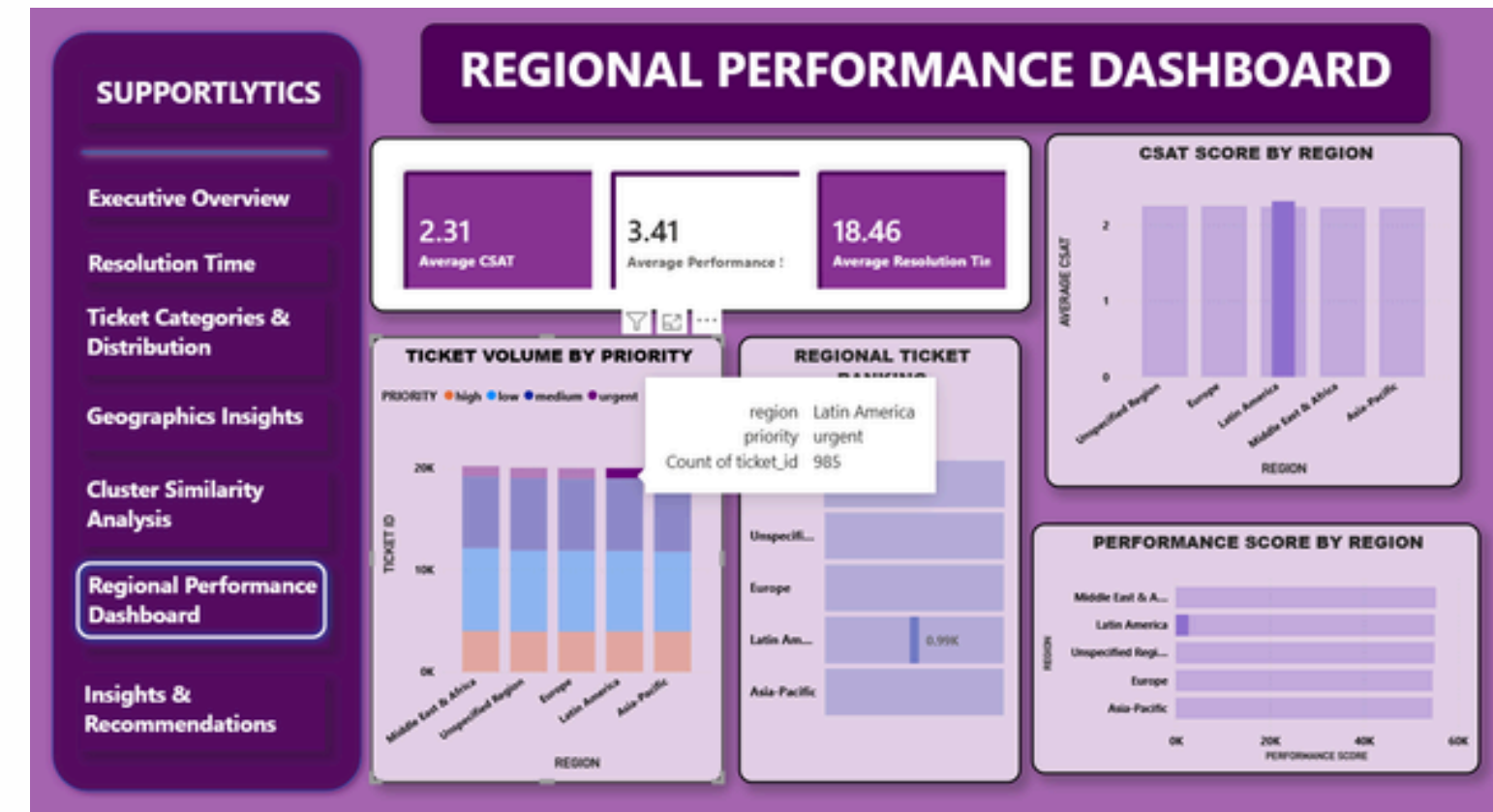
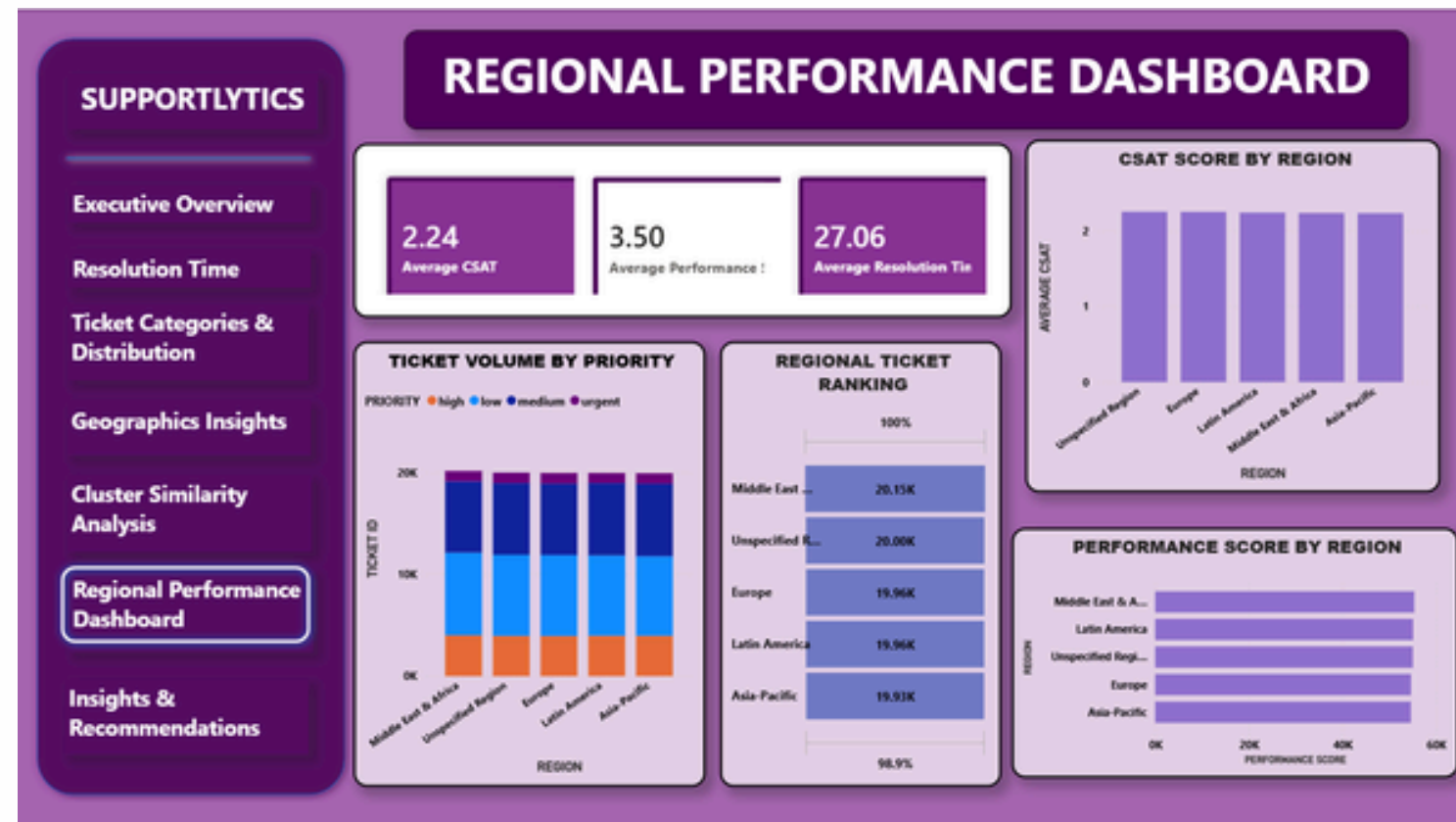
Key Insights

- Cluster size vs performance score
- Similarity score comparison
- Regional performance comparison
- Customer satisfaction across regions
- Resolution time and performance relationship

Visuals Used

- Scatter Plot
- Bar Chart
- Column Chart
- Decomposition Tree
- Performance Summary Table





Insights & Recommendations

Objective



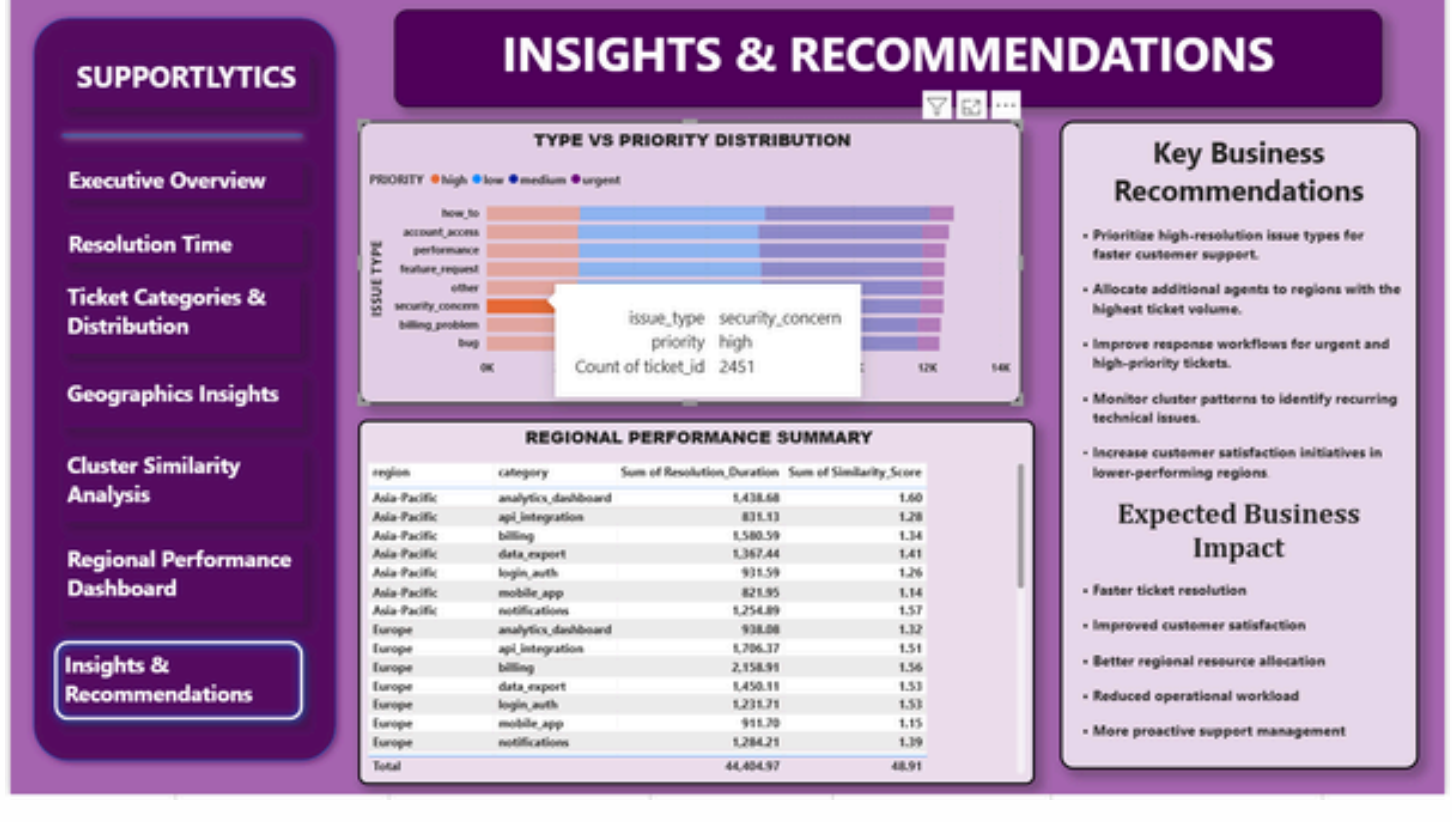
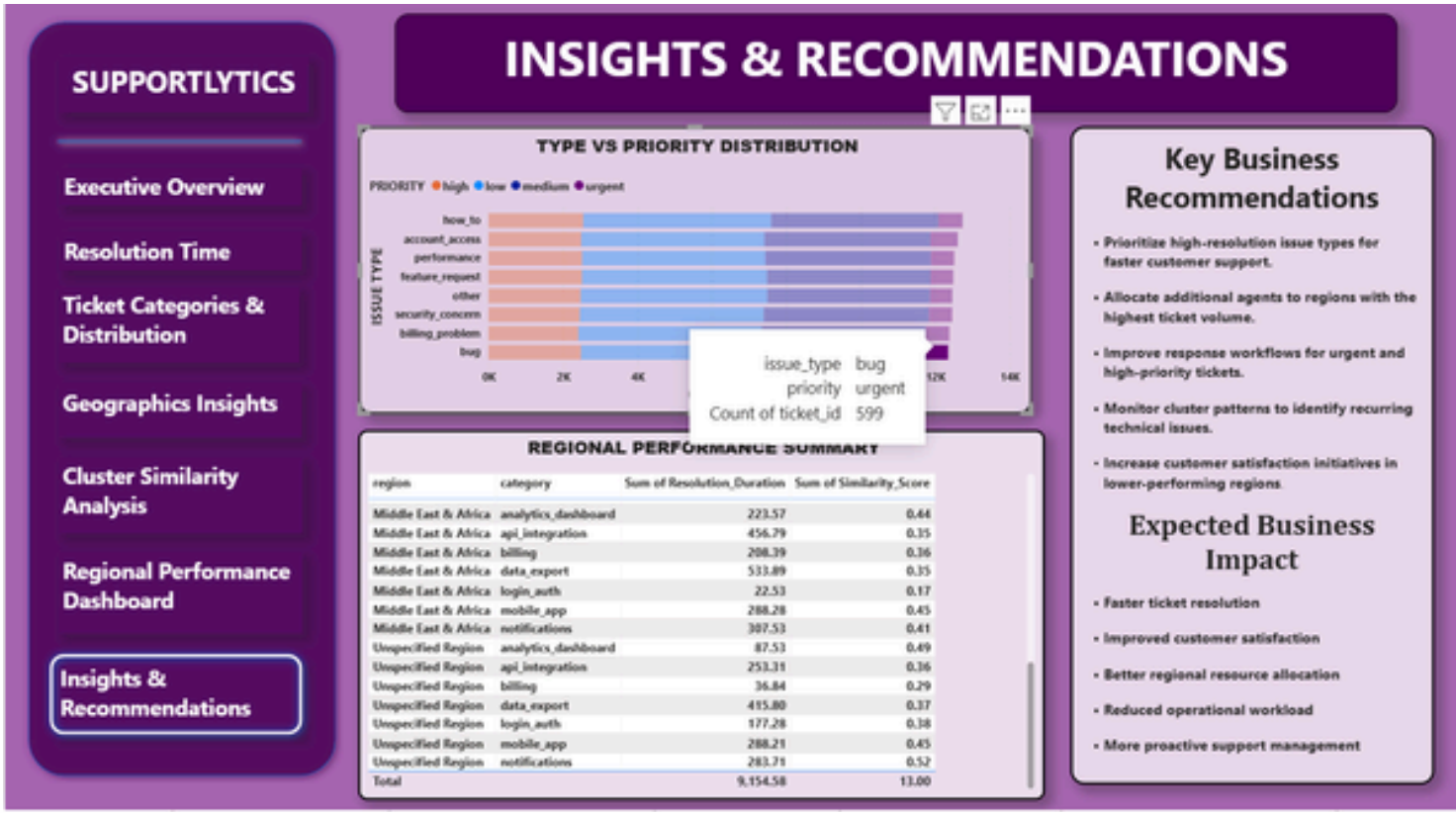
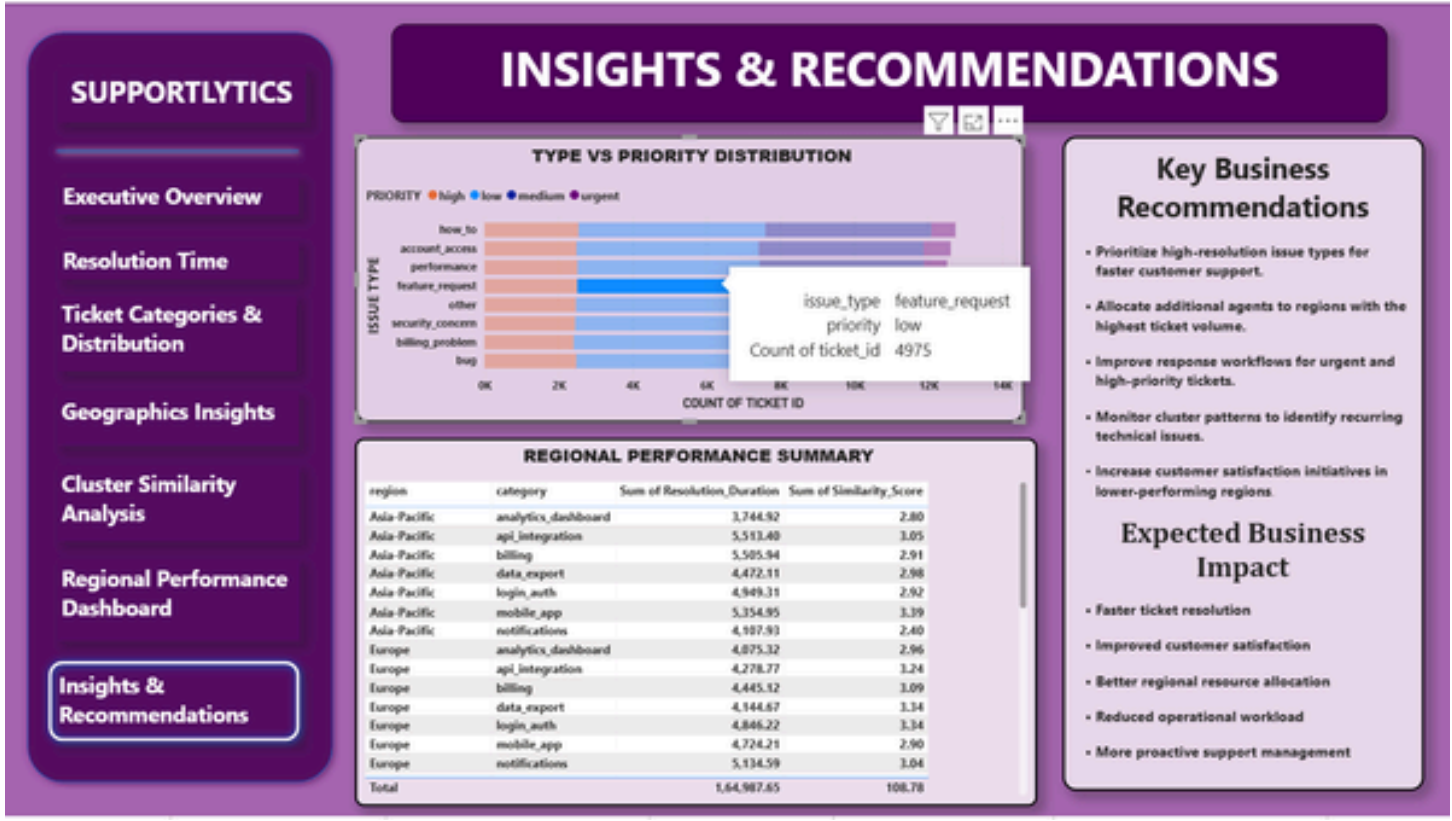
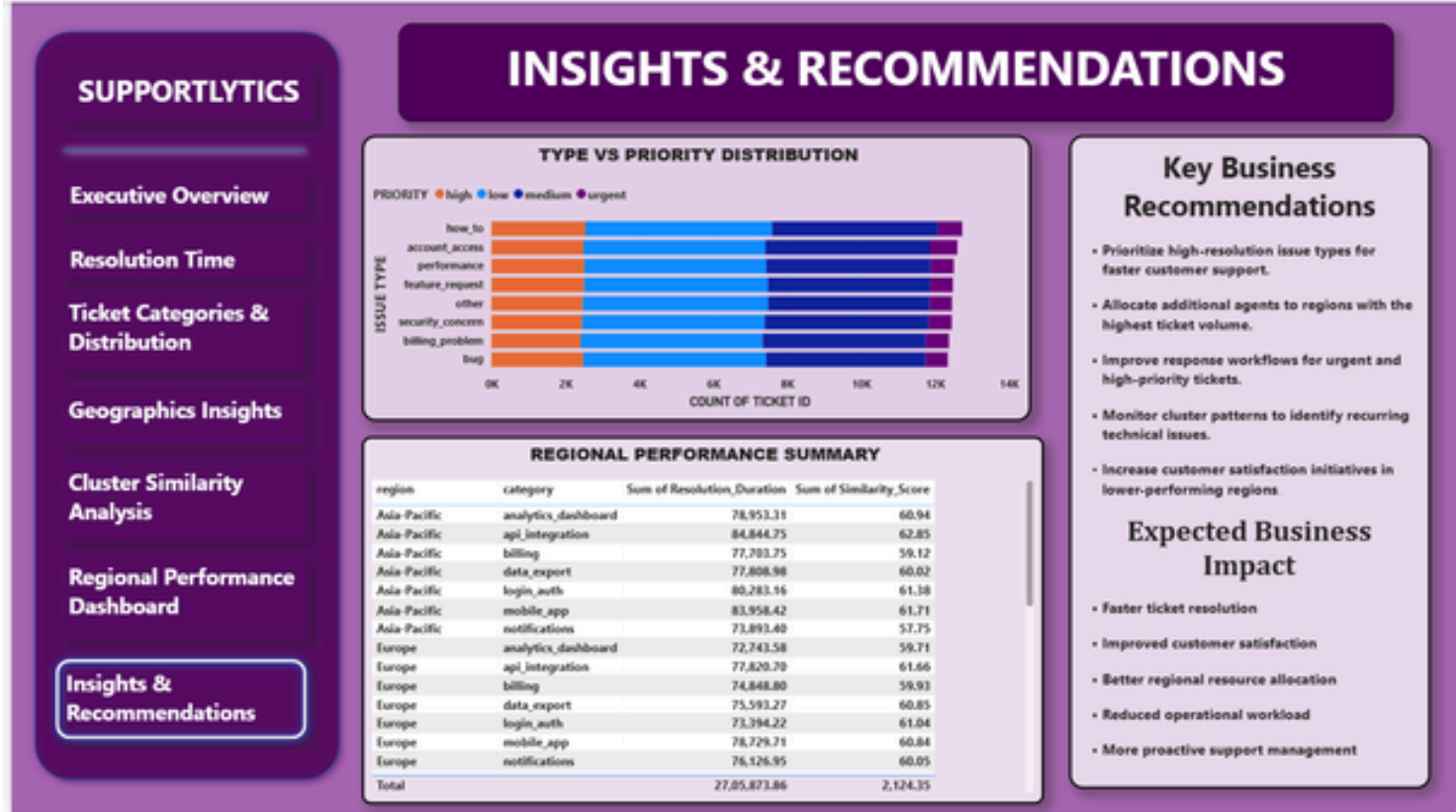
To summarize the most important findings and provide actionable recommendations.

Major Findings

- Some ticket categories generate significantly higher support volume.
- Resolution time varies across different regions and issue types.
- Certain regions require additional support resources.
- Cluster analysis identifies performance differences among ticket groups.

Recommendations

- Allocate resources to high-volume regions.
- Improve handling of long-resolution categories.
- Monitor low-performing clusters regularly.
- Optimize support queue distribution.



THANK YOU